



Feature Learning for Multimodal AI

From Foundations to Applications

Thanh-Sach LE

LTSACH@hcmut.edu.vn

Faculty of Computer Science and Engineering (CSE)

Ho Chi Minh City University of Technology (HCMUT)

HCMC Vietnam National University (HCM-VNU)

August 2025



Outline

- ▶ AI: Introduction
- ▶ Traditional AI
- ▶ Learning Features: CNN, RNN, GNN
- ▶ Learning Features: Transformer, Mamba
- ▶ Applications of Feature Learning in Multimodal AI
 - Spatial Intelligence Systems
 - Text-Based Human Search
- ▶ The Future of Feature Learning
- ▶ Conclusion

AI: Introduction

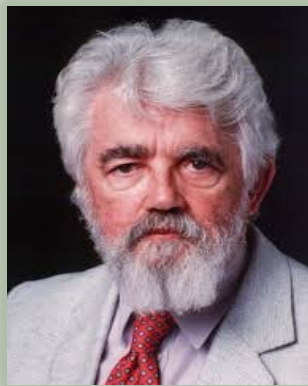
- What?
- Why?
- How?
- Current Capabilities
- Milestones



Artificial Intelligence: What?

- ▶ AI: a field of research aimed at enabling machines to perform **human-like abilities**.

First proposed at
the **Dartmouth Conference in 1956**

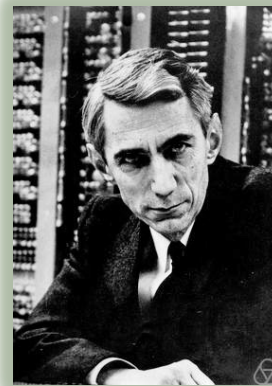


John McCarthy



Marvin Minsky

...



Claude Shannon



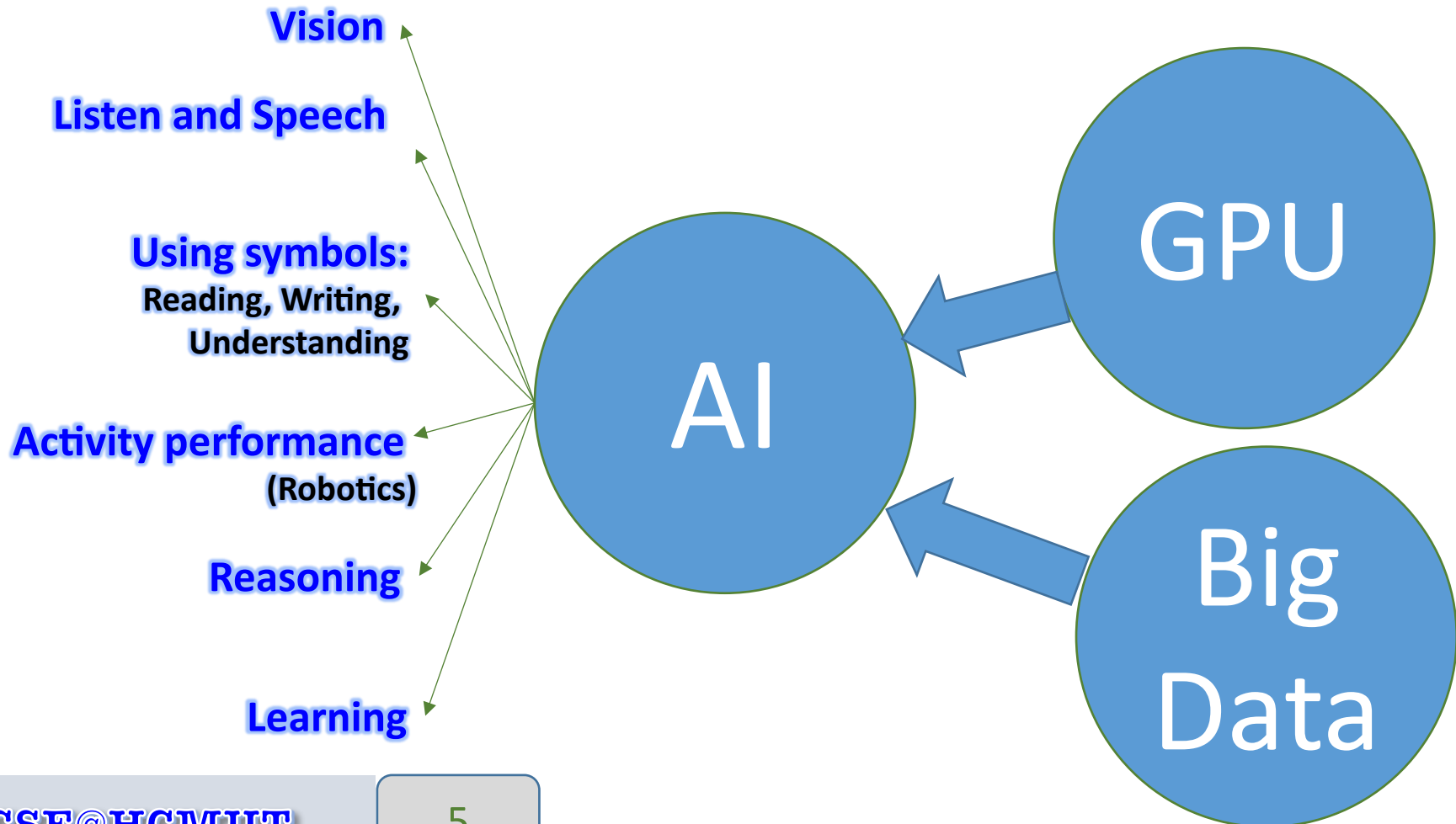
Allen Newell



Artificial Intelligence: What?

Human-like abilities

Supporting factors





Artificial Intelligence: Why?



To analyze big-data: with human resources: **infeasible!**
=> Use artificial intelligence to **automate** the analytics

To enable artificial intelligence:
=> Require big-data to **train** it



Artificial Intelligence: How?

Data, from:

- Social Networks
- Searching Texts
- Comment, feedback
- Location (GPS)
- Cameras, etc

Data, from:

- Other sources

Artificial Intelligence
(Learning Techniques)

GPU:



Interest capability
Purchasing power
Age of audience
Intent of audience
Customer preferences
(Many more)



Artificial Intelligence:

How?

AI: A mathematical function to map from inputs to outputs

Data, from:

- Social Networks
- Searching Texts
- Comment, feedback
- Location (GPS)
- Cameras, etc

Data, from:

- Other sources

Artificial Intelligence
(Learning Techniques)

GPU:



Interest capability
Purchasing power
Age of audience
Intent of audience
Customer preferences
(Many more)



Artificial Intelligence:

Current capabilities

Vision

1

Listen and Speech

2

Using symbols:

Reading, Writing,
Understanding

3

Activity performance

(Robotics)

4

Reasoning

5

Learning

6

Core tasks:

Detection of objects: 9/10

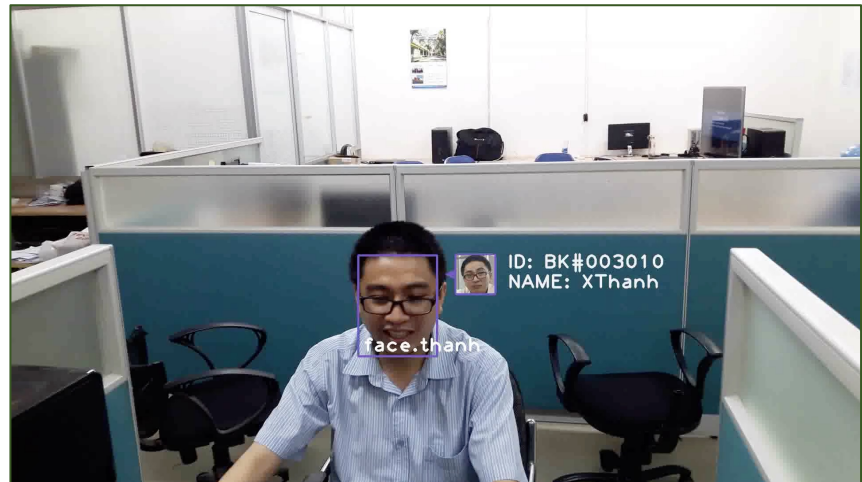
Tracking of objects: 7/10 (slows)

Identification: 9/10

Segmentation: 8/10

Description: 8/10 (data: too-big)

Others: ?





Artificial Intelligence:

Current capabilities

Vision

1

Listen and Speech

2

Using symbols:

Reading, Writing,
Understanding

3

Activity performance

(Robotics)

4

Reasoning

5

Learning

6

Core tasks:

Detection of objects: 9/10

Tracking of objects: 7/10 (slows)

Identification: 9/10

Segmentation: 8/10

Description: 8/10 (data: too-big)

Others: ?





Artificial Intelligence:

Current capabilities

Vision

1

Listen and Speech

2

Using symbols:

Reading, Writing,
Understanding

3

Activity performance
(Robotics)

4

Reasoning

5

Learning

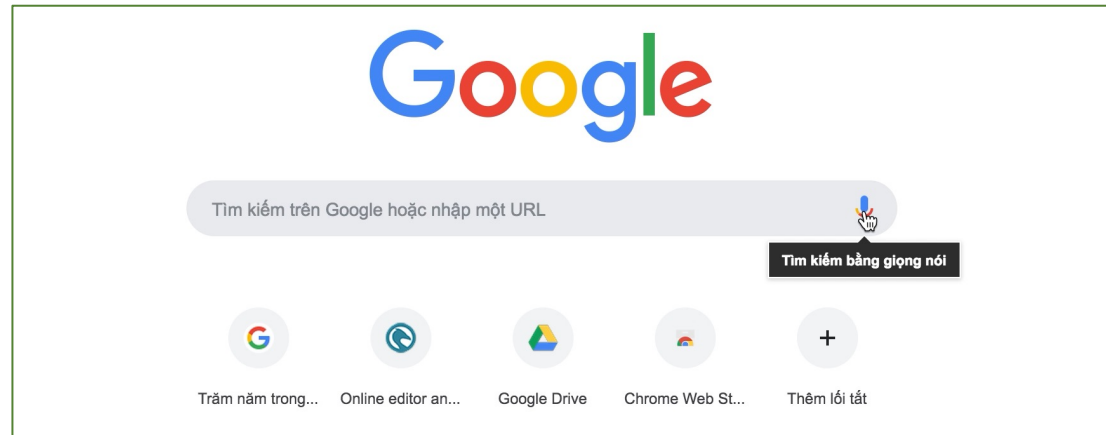
6

Core tasks:

Speech2Text: 9/10

Text2Speech: 9/10

Speech Segmentation: 9/10





Artificial Intelligence:

Current capabilities

Vision

1

Listen and Speech

2

Using symbols:

Reading, Writing,
Understanding

3

Activity performance

(Robotics)

4

Reasoning

5

Learning

6

Core tasks:

ChatGPT v4: 9

(Writing is better than me!,
even in Vietnamese 😊)

(Live Demo)



Artificial Intelligence:

Current capabilities

Vision

1

Listen and Speech

2

Using symbols:

Reading, Writing,
Understanding

3

Activity performance
(Robotics)

4

Being soon with Autonomous vehicles and
Human-like Robots (from Tesla)

Reasoning

5

The weakest capability currently

Learning

6

Machine learning/Deep learning:
Fine but require lots of GPUs



Milestones of AI and Feature Learning

► 1956 – 2012: Traditional AI

- Handcrafted features (SIFT, TF-IDF, etc.)
- Classical machine learning (SVM, Naive Bayes, RF...)

► 2012 – 2017: Deep Learning Emergence

- CNNs for vision (AlexNet, ResNet, etc.)
- RNN, LSTM, GRU for sequences
- Rise of domain-specific backbones



Milestones of AI and Feature Learning

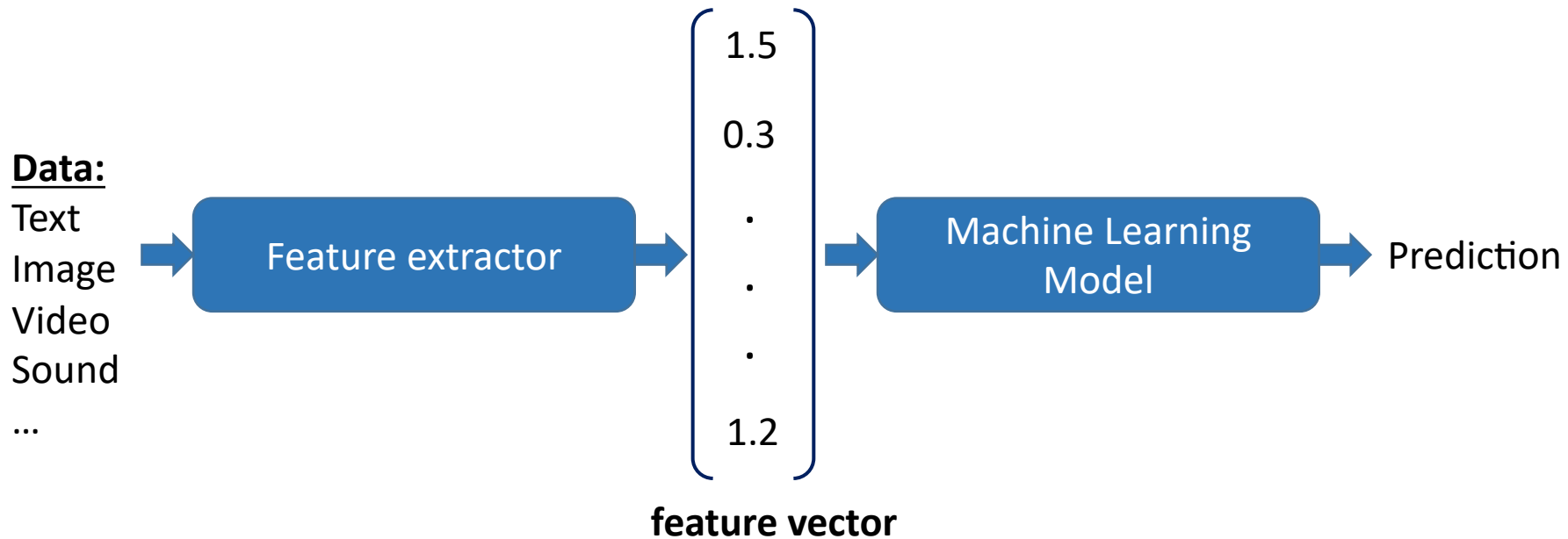
- ▶ **2017 – ~2022: Attention and Unification**
 - Transformer for text (BERT, GPT...)
 - ViT for vision, CLIP for multimodal
 - Foundation Models emerge
- ▶ **~2022 – Now: Multimodal AI Era**
 - Vision–Language models (e.g. Flamingo, BLIP, GPT-4V)
 - Spatial + language reasoning
 - Unified feature learning across modalities

Traditional AI

Handcrafted Features and Traditional Machine Learning



Traditional AI





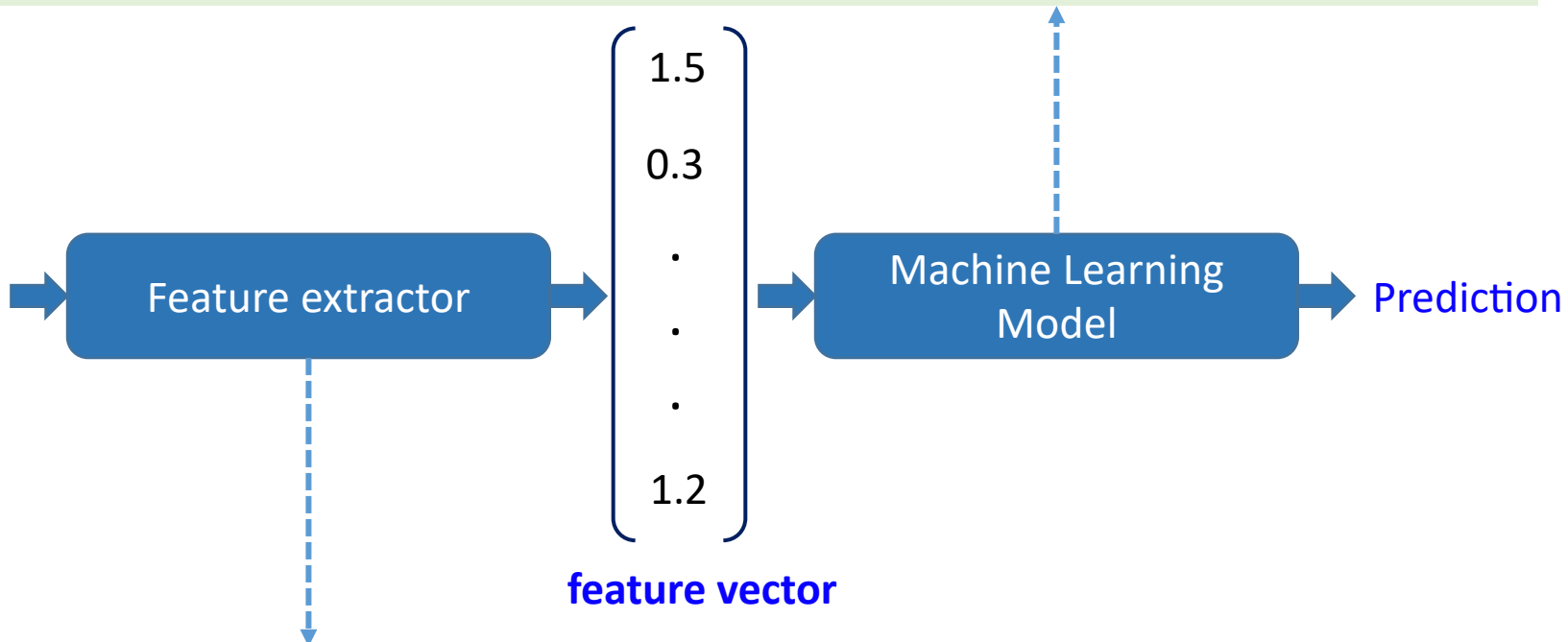
Traditional AI

Machine Learning Model:

A learnable program that takes feature vectors to learn patterns and make predictions.

Data:

Text
Image
Video
Sound
...

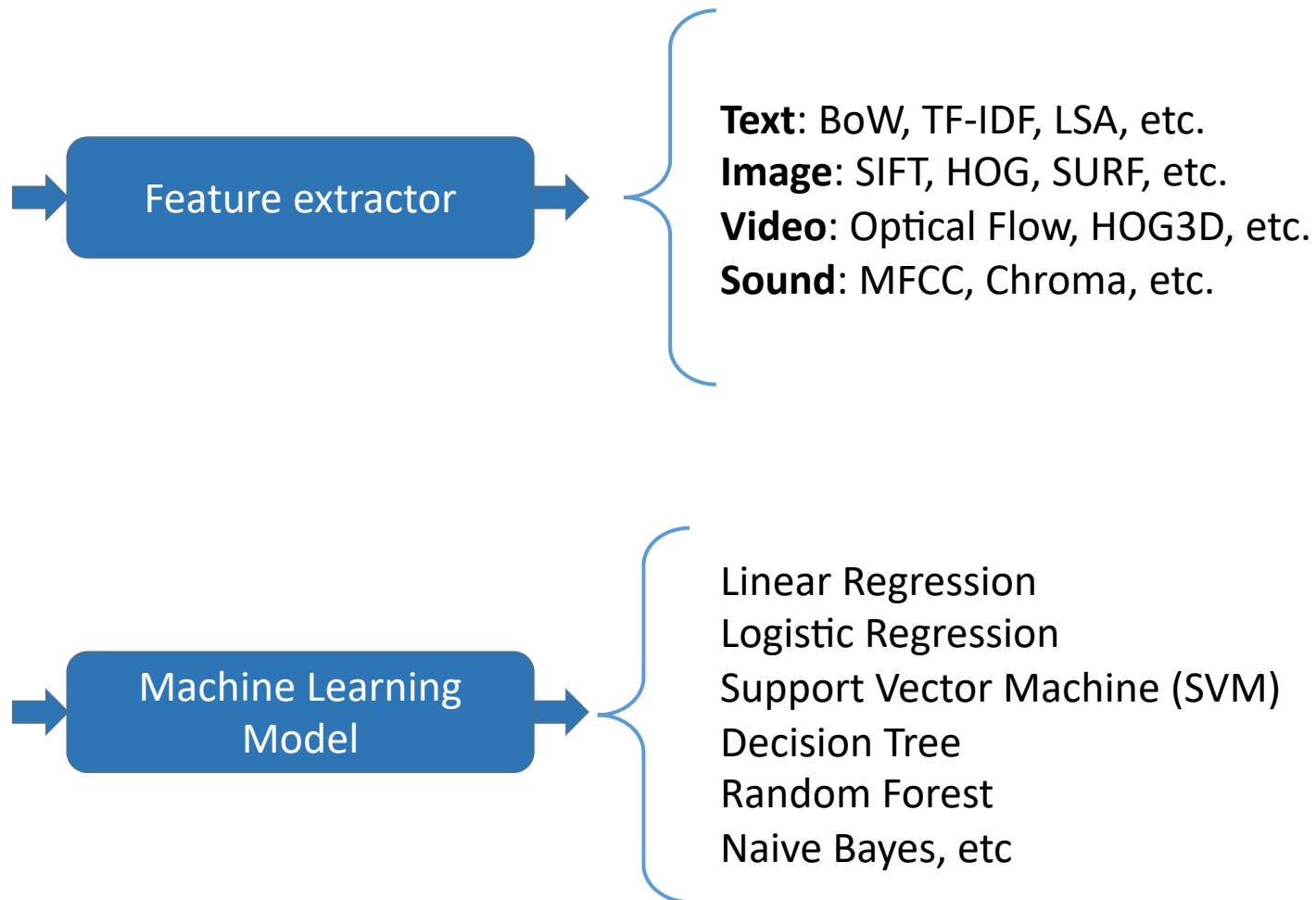


Feature extractor:

A special algorithm that processes a data and represents it as a numerical vector.

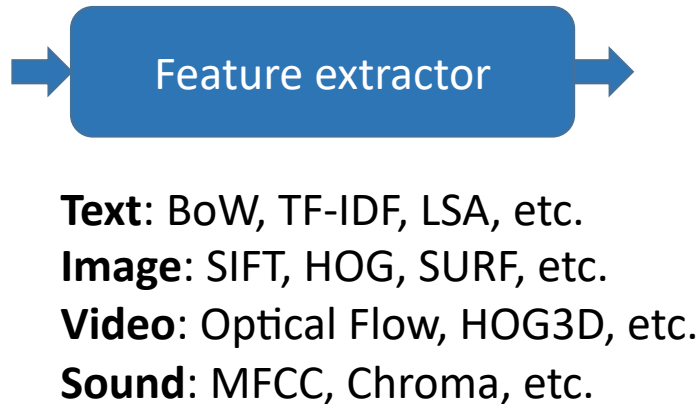


Traditional AI

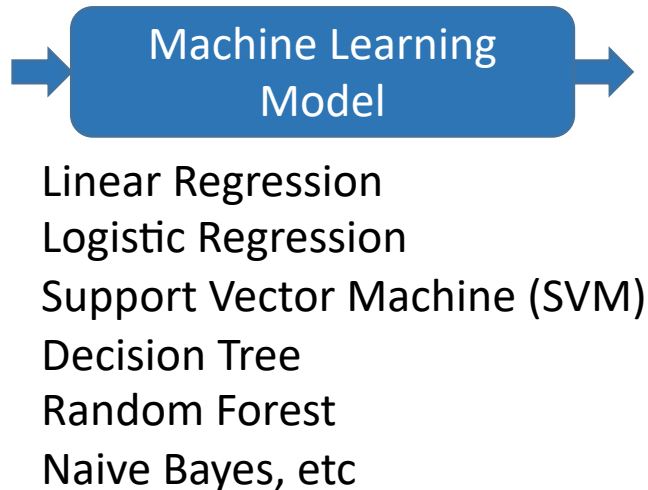




Traditional AI



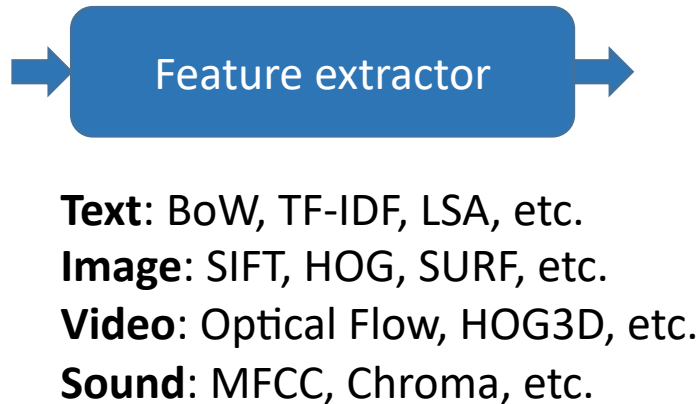
Separate algorithms and courses for CV, NLP, Speech, etc.



Many different algorithms for learning patterns from feature vectors.



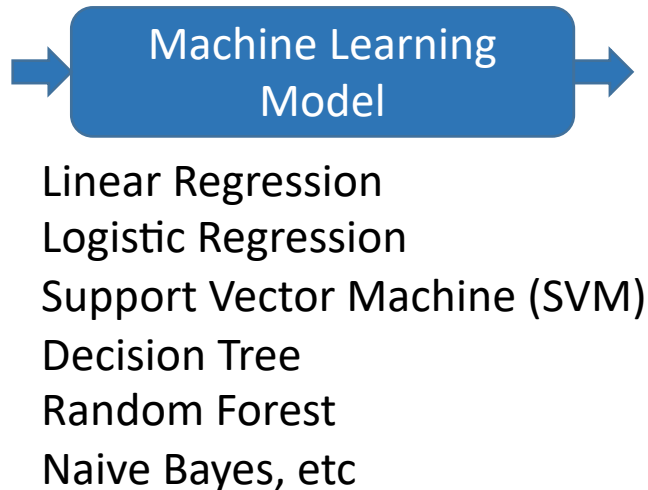
Traditional AI



Separate algorithms and courses for CV, NLP, Speech, etc.



Causing difficulties



Many different algorithms for learning patterns from feature vectors.



Traditional AI

- ▶ Handcrafted features **require** expert design for each domain.
- ▶ Algorithms **require** domain-specific design and are non-adaptive to new data.
- ▶ Many separate algorithms **require** development for different modalities.
- ▶ Large-scale deployment **requires** overcoming limited scalability.

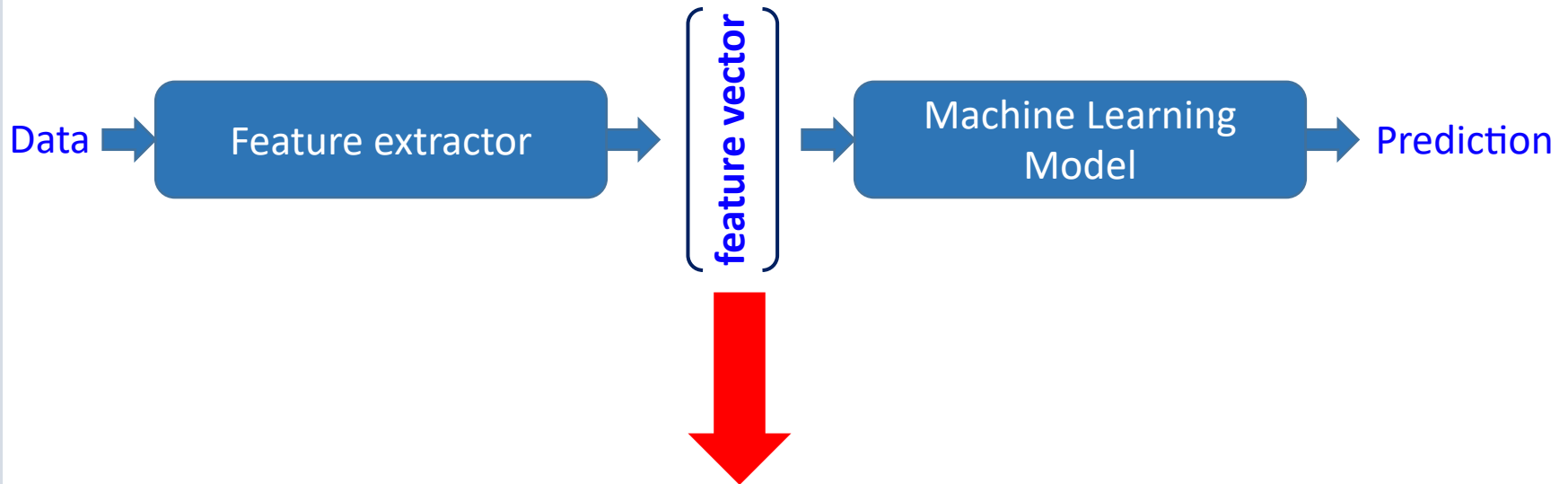
Learning Features: CNN

- From Handcrafted to Learned Features
- The Beginning of the deep learning era, AlexNet
- **Feature Learning + Unified Architecture**
- **Training Algorithms**
- **Data**
- **GPU**



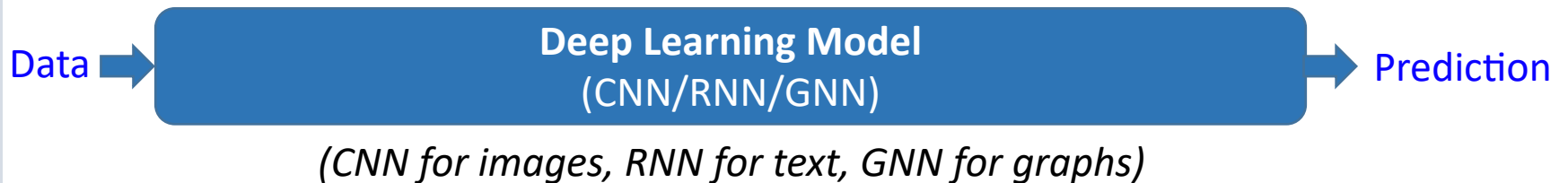
From Handcrafted to Learned Features

Traditional machine learning:



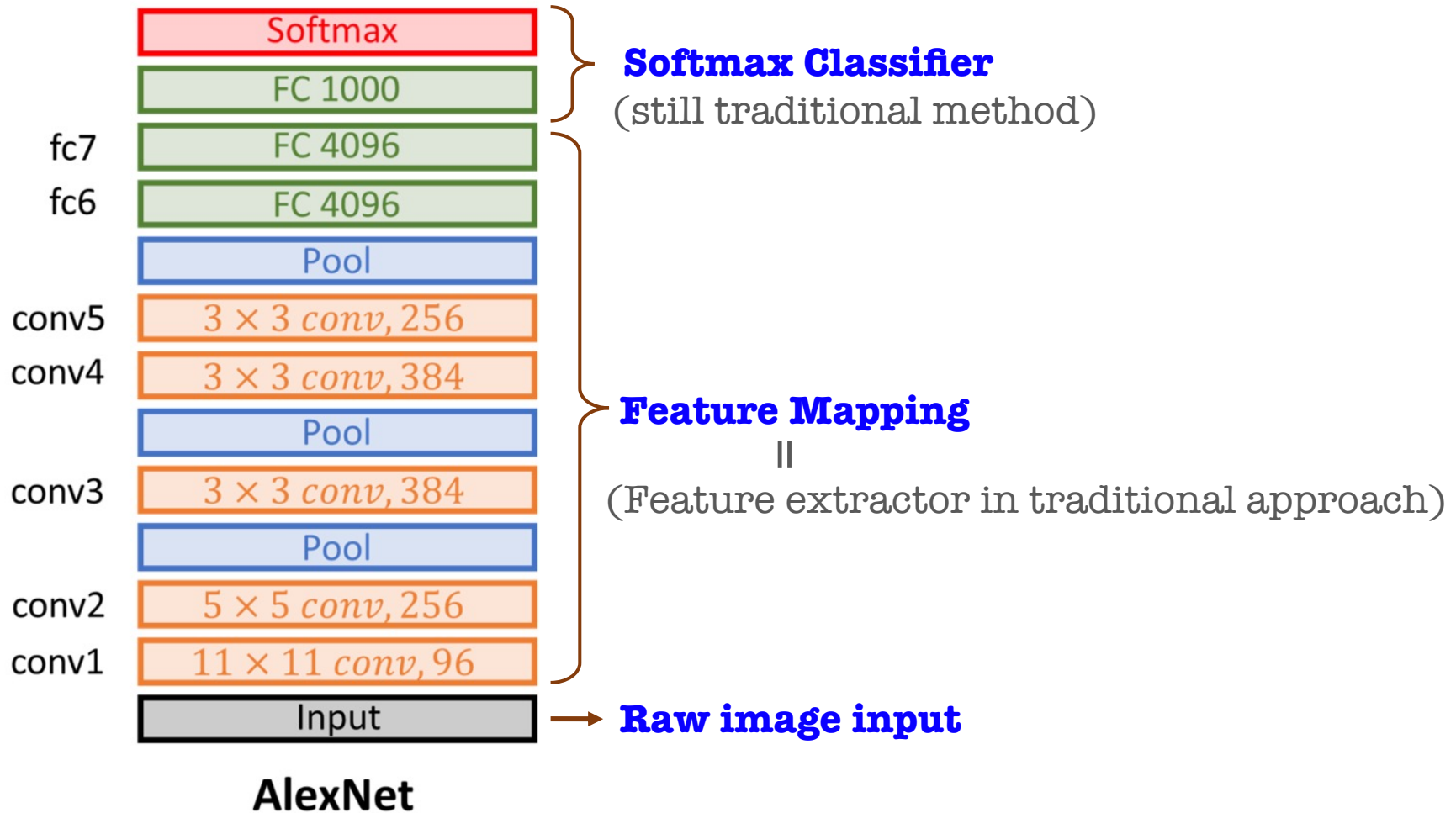
Paradigm Shift (from 2012):

Deep Learning learns features and makes predictions directly from raw data.





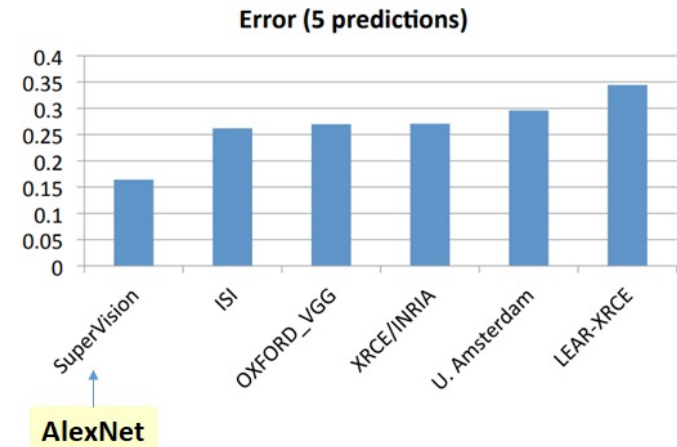
The Breakthrough – AlexNet (2012)



IMAGENET



Ranking of the best results from each team



1. ImageNet 2012 Champion

- ✓ Top-5 error: **15.3%** vs **26.2%** (second place)
- ✓ Unprecedented winning margin in ImageNet history.

2. GPU Acceleration

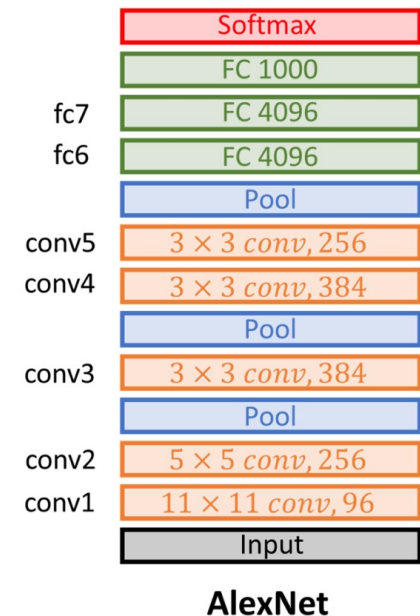
- ✓ Trained on 2× NVIDIA GTX 580, about 10× faster than CPU.

3. Key Innovations

- ✓ **ReLU** activation → faster training than sigmoid/tanh.
- ✓ **Dropout** → reduced overfitting.
- ✓ **Data augmentation** → increased data diversity.

4. Impact

- ✓ **Marked the beginning of the Deep Learning era** 🚀
- ✓ Inspired research on deep networks for vision, NLP, and beyond.





Why CNN Took Off?

1. Feature Learning + Unified Architecture

- Learns features automatically, no hand-crafted design
- Local connectivity, weight sharing, translation invariance

2. Training Algorithms

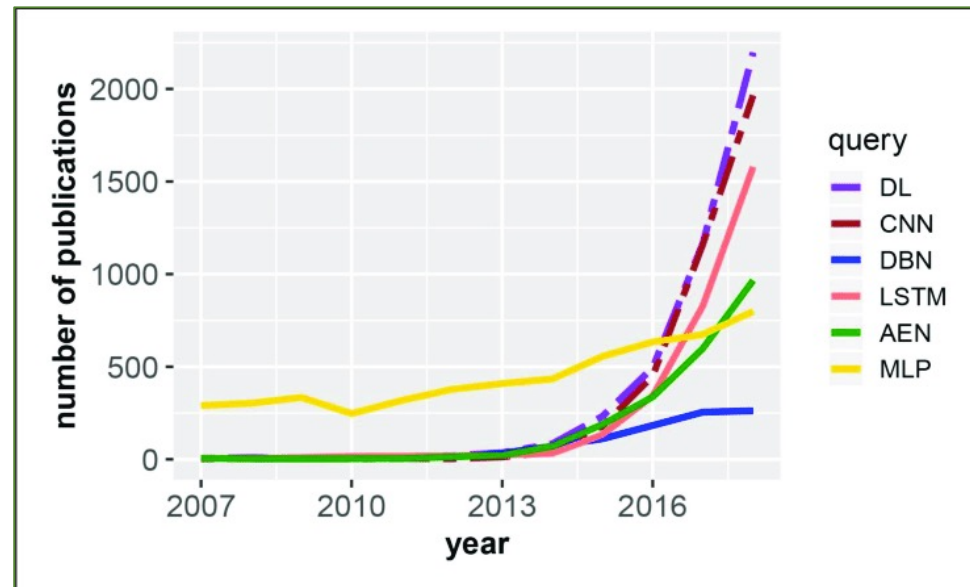
- Backprop + mini-batch SGD
- Regularization (Dropout, BatchNorm), ReLU

3. Data

- Large-scale datasets (e.g., ImageNet)
- Data augmentation

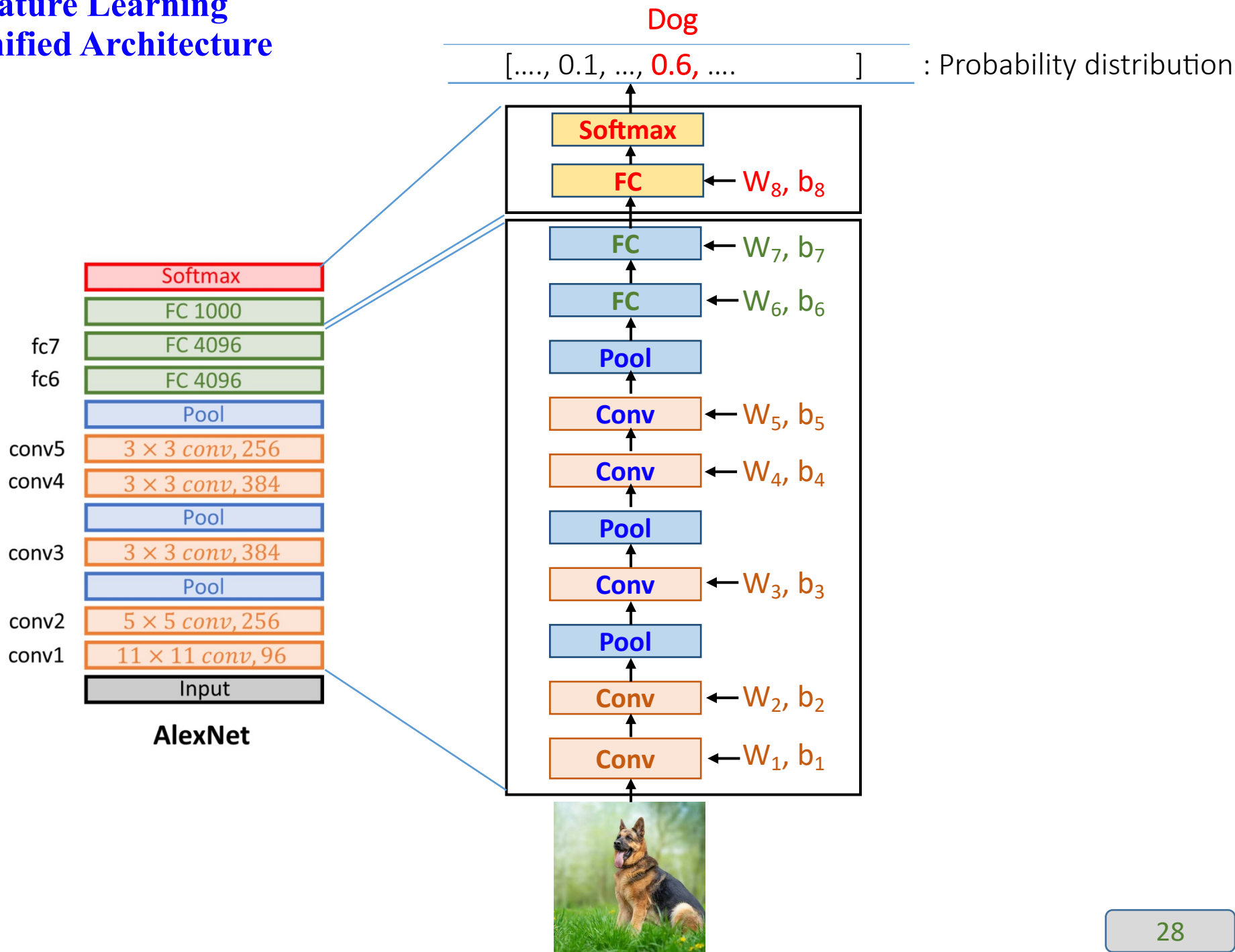
4. GPU

- Parallel processing, $\sim 10\times$ faster training
- Ideal for conv/matrix ops

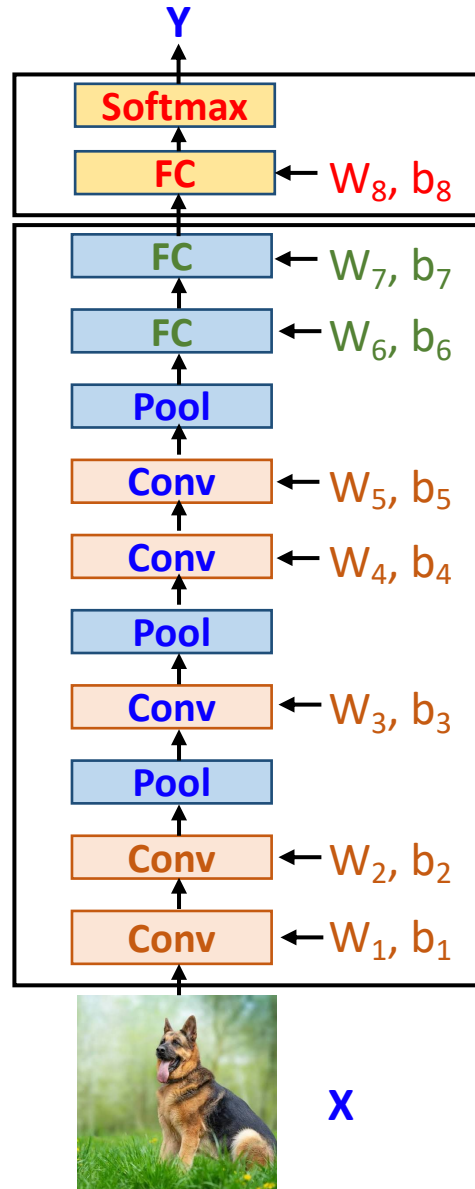


Web of Science (Emmert-Streib et al., 2020)

Feature Learning Unified Architecture



Feature Learning and Unified Architecture



(1) Model = computational graph

- Vertex: operator
- Edge: flow of data

⇒ Standard method for representing models

(2) Model = Math. Function

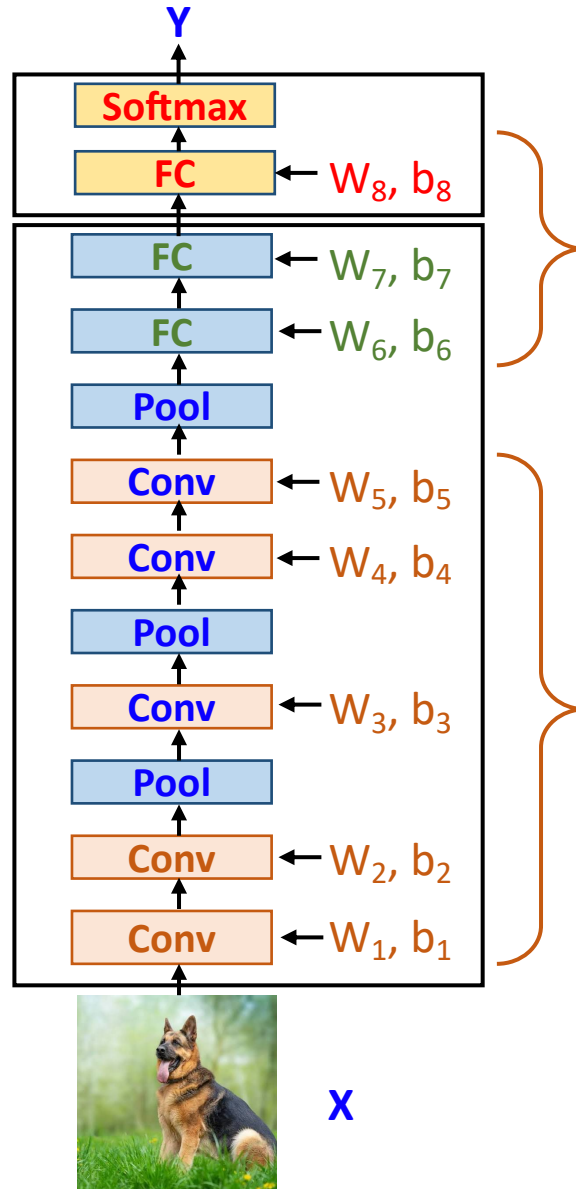
- Variable = learnable weights
- Input X = input data
- Output Y = Model's prediction

$$Y = F(W_1, b_1, \dots, W_8, b_8; X)$$



W_i, b_i : **learnable**
From pairs: $(X \rightarrow Y)$

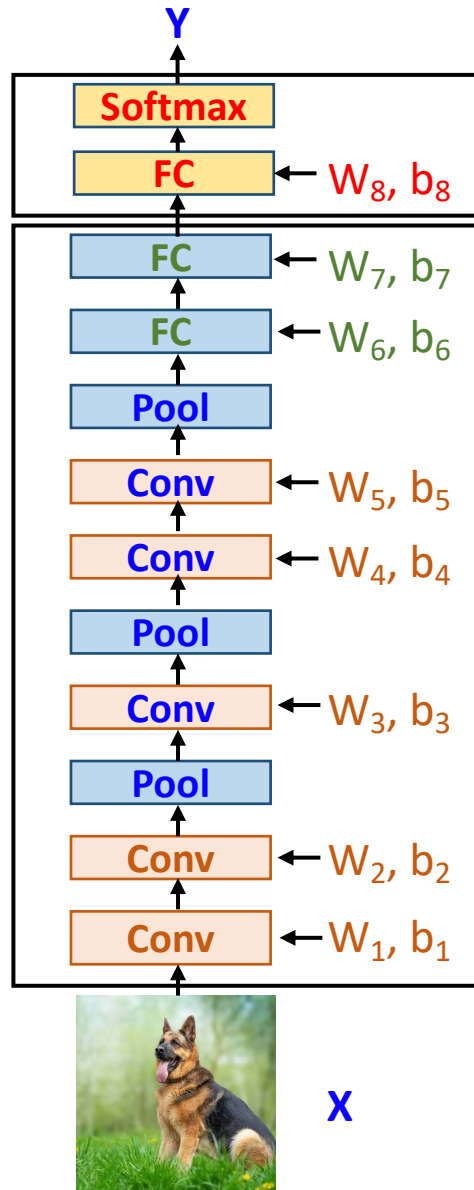
Feature Learning and Unified Architecture



Conv = Convolution

- Pre-2012, W_i, b_i hand-designed
- Post-2012: learnable parameters
- Single Conv: Locality
- Stack of Convs: local $\rightarrow$ global
- Locality + shared weights $\rightarrow$ fewer parameters (vs. FC)

Feature Learning and Unified Architecture

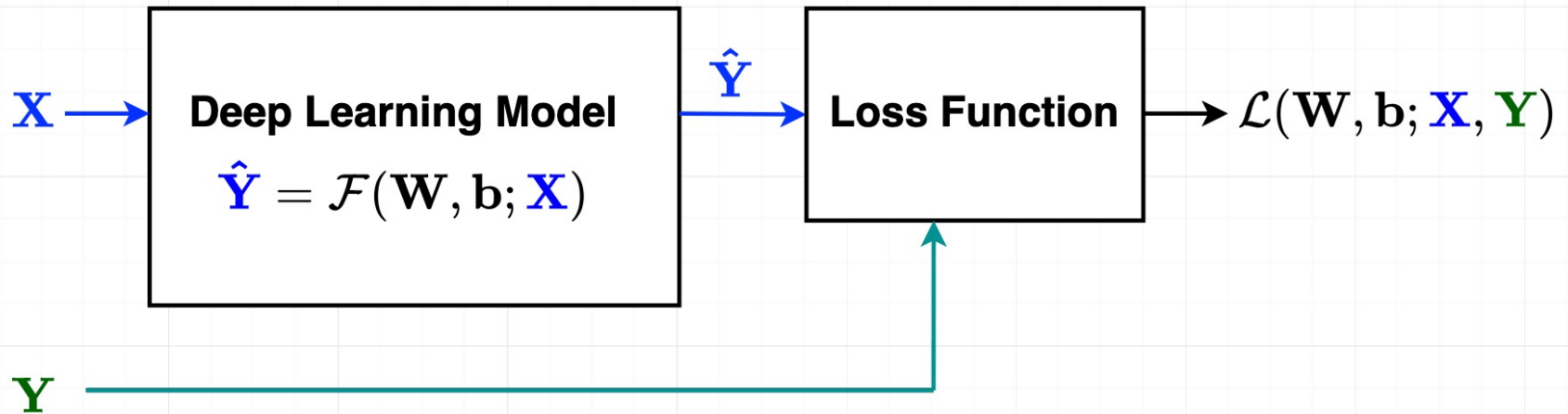


To solve a problem:

- (1) Construction model (graph)
- (2) Collect data (X) and give it label (Y)
- (3) Train/Val/Test the model
- (4) Deploy the model



Training algorithms



Training = Optimization:

$$\mathbf{W}^*, \mathbf{b}^* = \underset{\mathbf{W}, \mathbf{b}}{\operatorname{argmin}} \mathcal{L}(\mathbf{W}, \mathbf{b}; \mathbf{X}, \mathbf{Y})$$

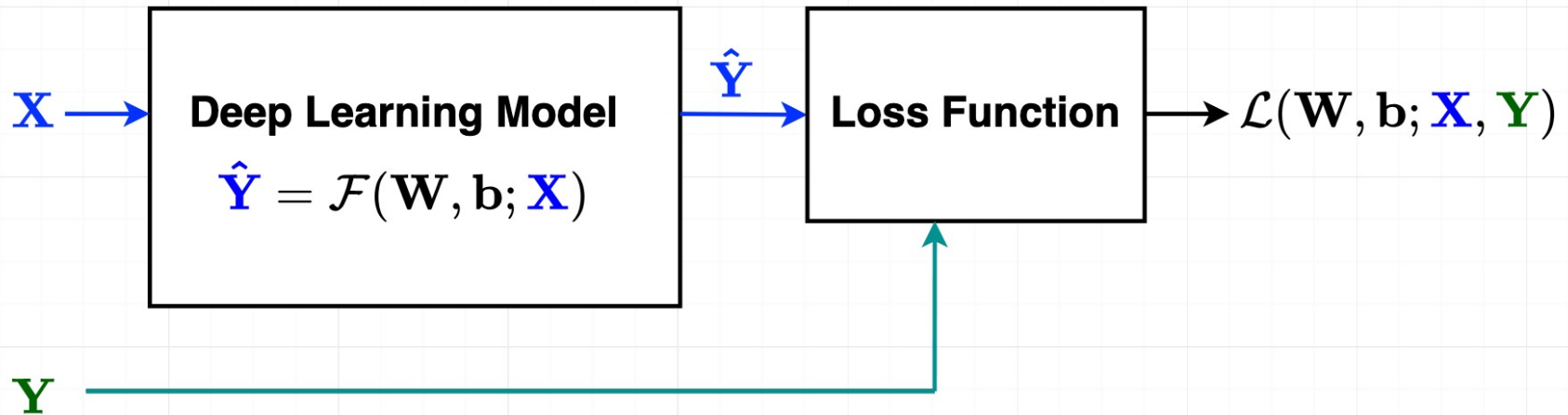


Algorithm: SGD, Adam, AdamW, ...

Framework: PyTorch, Keras, Huggingface, ...



Training algorithms



Training = Optimization:

$$\mathbf{W}^*, \mathbf{b}^* = \underset{\mathbf{W}, \mathbf{b}}{\operatorname{argmin}} \mathcal{L}(\mathbf{W}, \mathbf{b}; \mathbf{X}, \mathbf{Y})$$



The most difficult part: computing the gradient $\left[\frac{\mathcal{L}}{\partial \mathbf{W}}, \frac{\mathcal{L}}{\partial \mathbf{b}} \right]$

Done automatically by frameworks:

+ **Pytorch**: `loss.backward()`

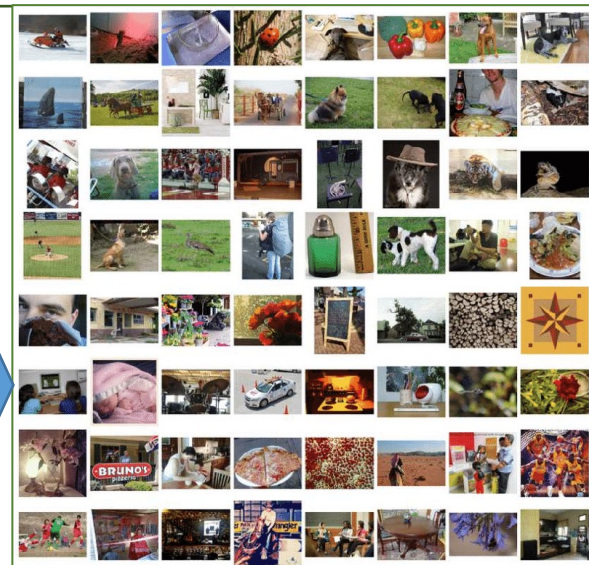
+ **TensorFlow/Keras**: `tf.GradientTape()`



Datasets



MNIST



ImageNet

airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck



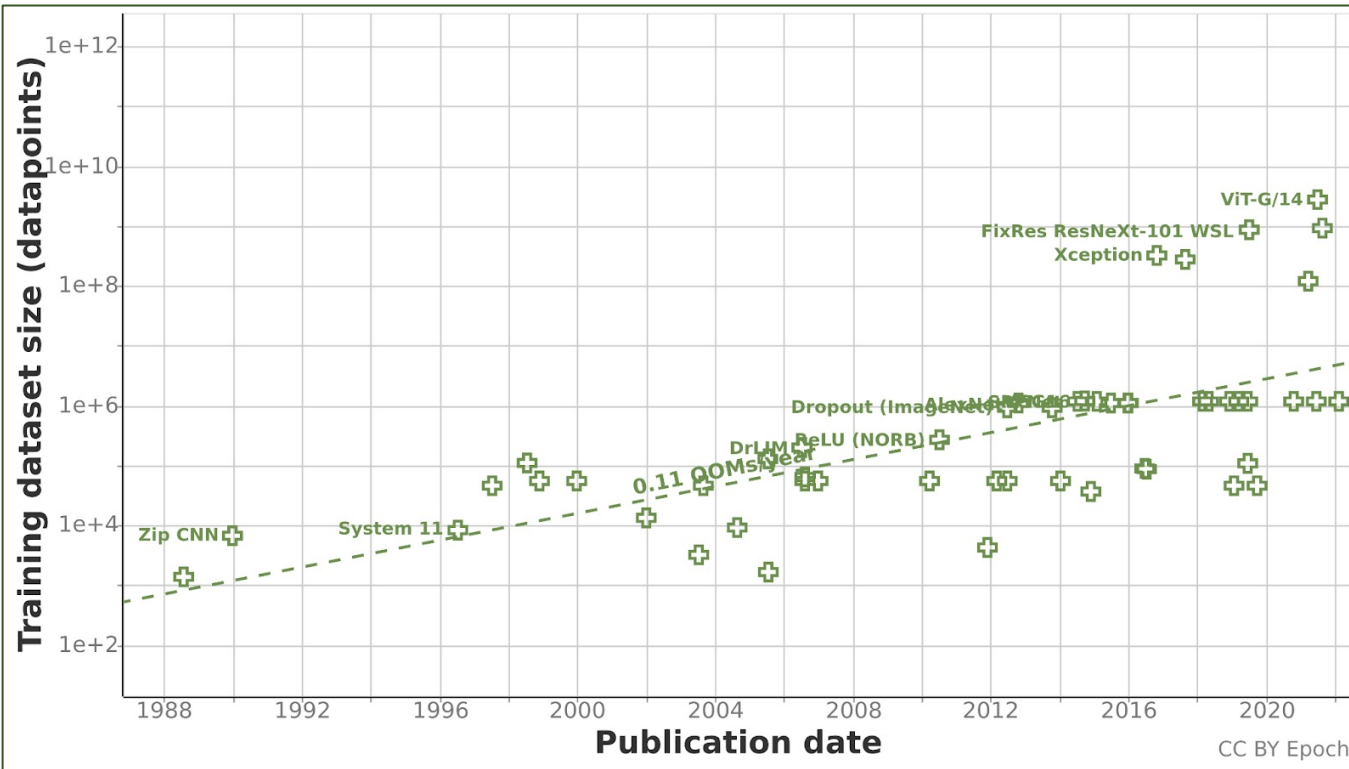
CIFAR10

Food2K





Datasets



Evolution of vision datasets

Most models: 60K–1M samples

Some: ~3B samples



Datasets

Summary of trends for each domain

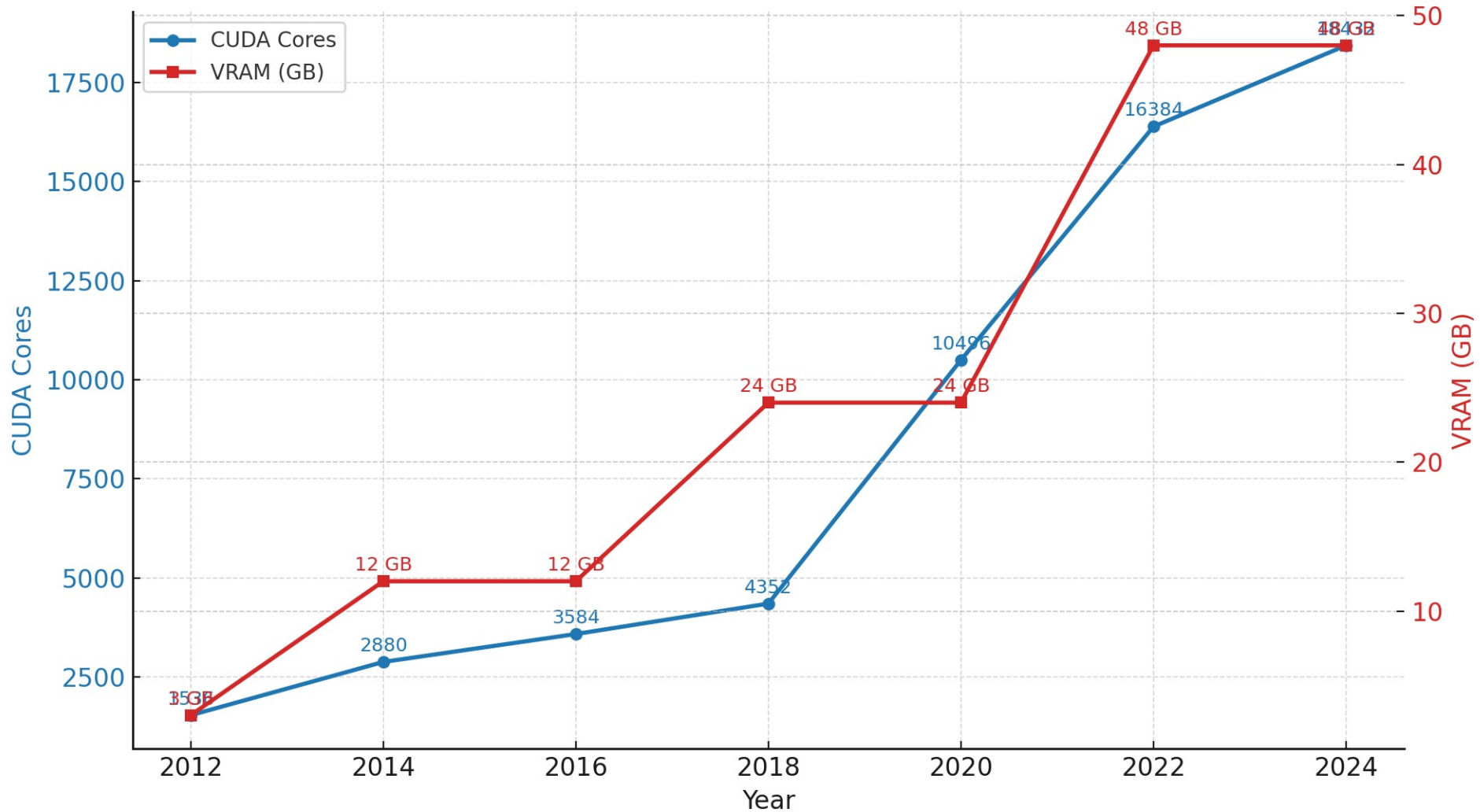
Domain	Scale (data points)	Yearly growth (OOMs/year)	Yearly growth (OOMs/year) (95% CI)	#systems
Language	1e2- 2e12	0.22	[0.18 ; 0.28]	79
Vision	2e3 - 3e9	0.09	[0.08 ; 0.11]	55
Speech	9e2 - 3e12	0.21	[0.17 ; 0.30]	13
Games	7e5 - 4e11	0.09	[0.08 ; 0.15]	12
Recommendation	1e8 - 1e10	0.05	[0.00 ; 0.47]	11
Drawing	6e4 - 4e9	0.43	[0.17 ; 0.64]	10



By 2024, dataset sizes had reached billions of samples for all types of data.



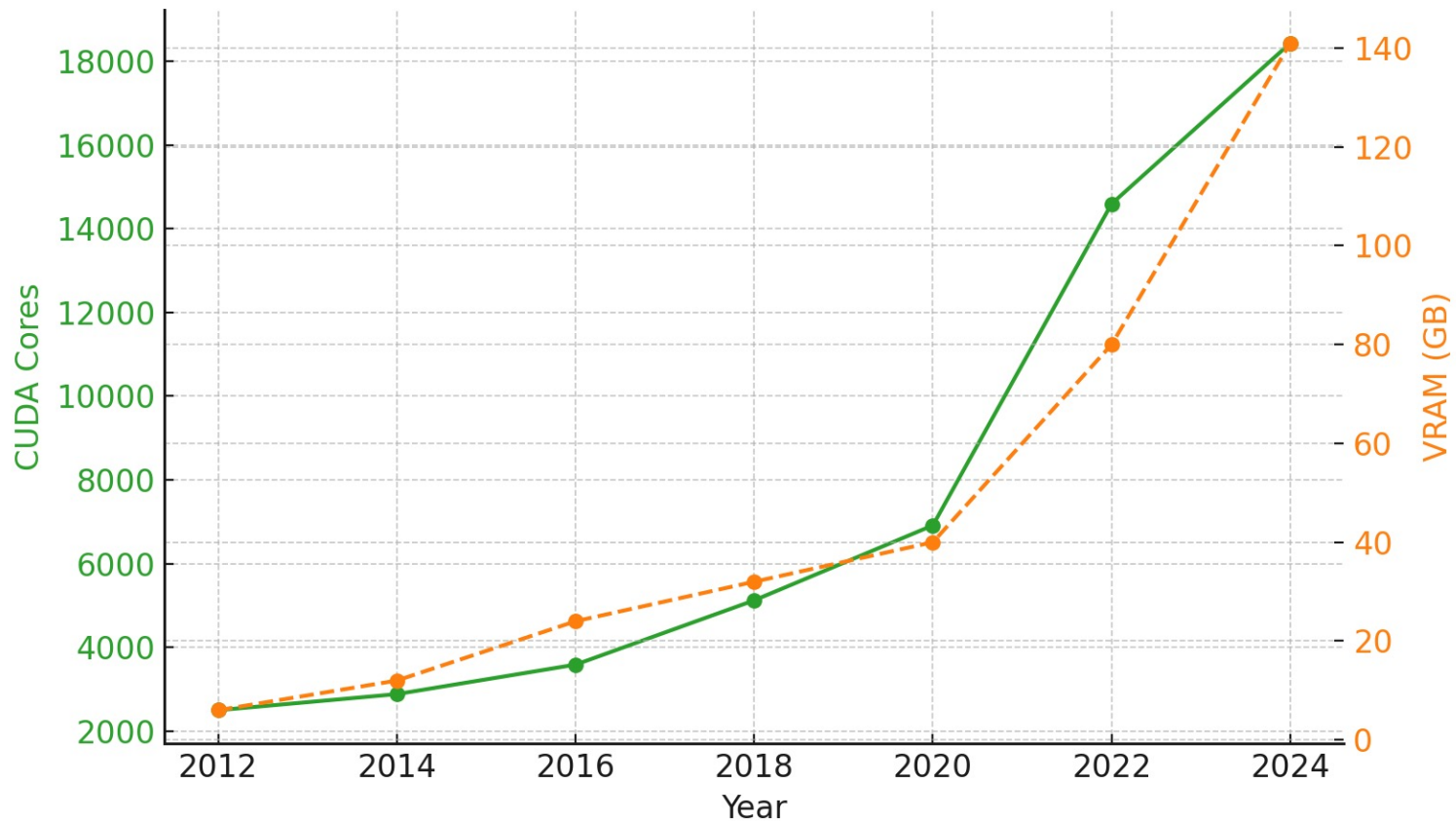
Top Desktop GPUs: CUDA Cores vs VRAM Over Time





GPU

High-End Server GPUs: CUDA Cores vs VRAM (2012-2024)

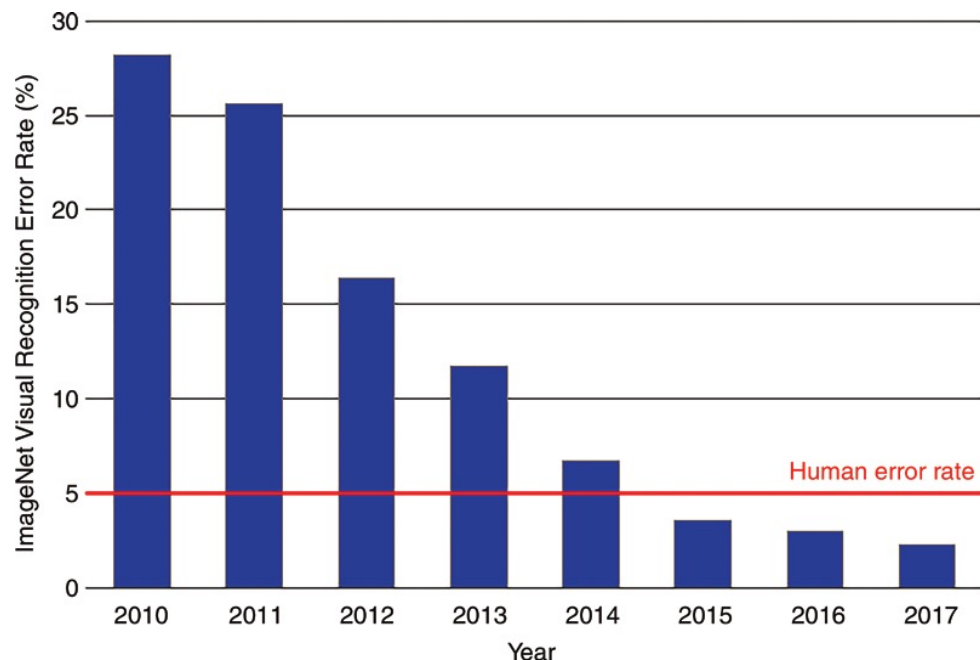




CNN – Results & Applications

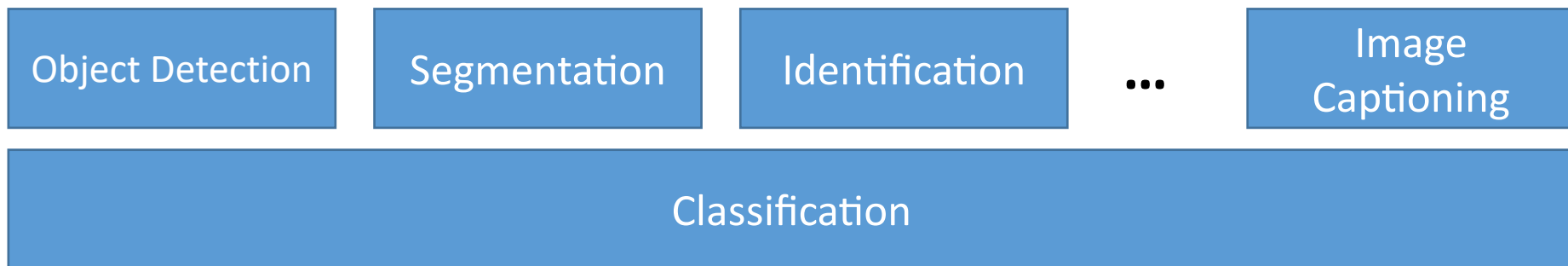
- **2012 (AlexNet):** Reduced Top-5 error from ~26% (traditional) to **15.3%** (ILSVRC).
- **2013 (ZFNet):** Improved visualization & tuning → **11.2% error**.
- **2014 (GoogLeNet, VGG):** Achieved **6.7% error**.
- **2015 (ResNet):** Only **3.6%**, surpassing human-level performance (~5%) on ImageNet.

➔ **2015 milestone:** CNNs officially outperformed humans in image classification.





Beyond Classification: CNN Applications



- **Object Detection:** R-CNN, Fast R-CNN, Faster R-CNN, SSD, Yolo, ...
- **Segmentation:** FCN, U-Net, Mask R-CNN, Unet-Like Models), Yolo5+, ...
- **Identification:** DeepFace, FaceNet, ArcFace, ...

Backbone CNNs from classification (VGG, ResNet, ...) are reused across most applications

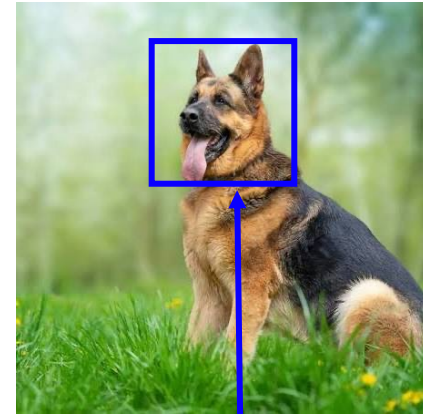
Learning Features: RNN

- From Spatial to Sequential
- Recurrent Neural Networks (RNNs)
- RNN $\rightarrow$ LSTM / GRU



From Spatial to Sequential

- CNNs excel at **spatial feature extraction** (images, grids).
- But many problems involve **sequences**:
 - Language (words, sentences)
 - Speech (sound waves)
 - Time series (finance, sensors)
- We need models that **capture temporal dependencies** → **RNNs**.



CNNs

excel at extracting
local spatial features (e.g., images),
but are weak at capturing
sequential dependencies
(e.g., language):

“The picture contains a dog”



Recurrent Neural Networks (RNNs)

- Designed for **sequential data**.
- Key idea:
 - Hidden state h_t carries information across time steps.
 - Can capture **temporal dependencies** (context/history).
- Applications:
 - Language modeling, machine translation.
 - Speech recognition.
 - Time-series forecasting.
- Limitation: **vanishing/exploding gradients** (long sequences).



RNN $\rightarrow$ LSTM / GRU

❖ Motivation:

- Standard RNN struggles with long-term dependencies.

❖ LSTM:

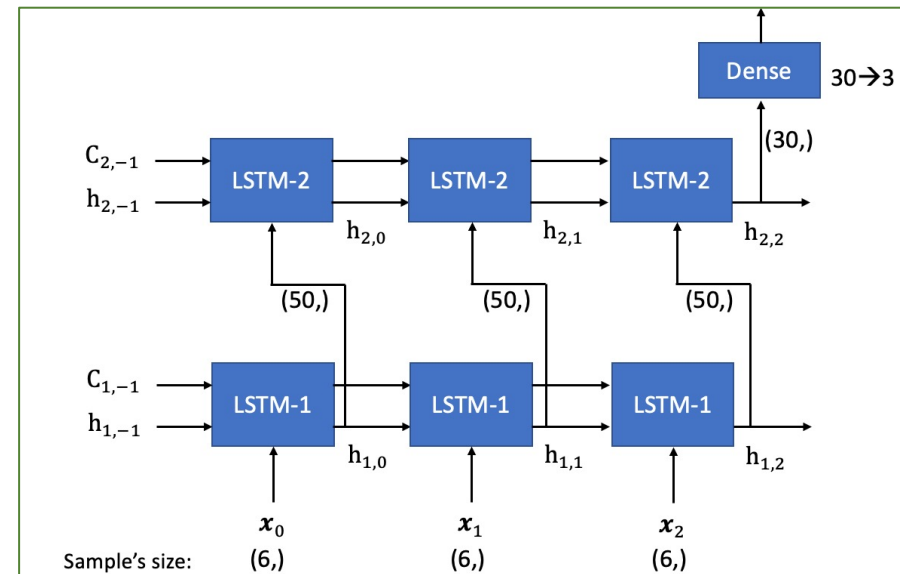
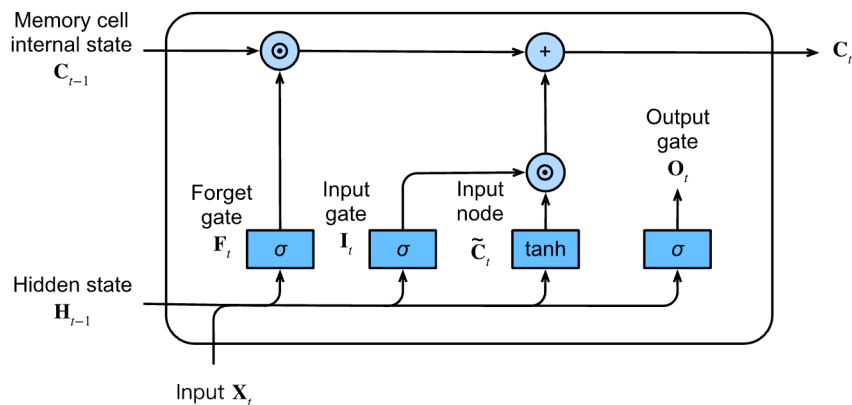
- Introduced memory cells and gates.

❖ GRU:

- Simplified LSTM, fewer parameters.

❖ Applications:

- NLP, speech, and seq2seq tasks.



Sample's size:

$(6,)$

$(6,)$

$(6,)$

σ FC layer with activation function

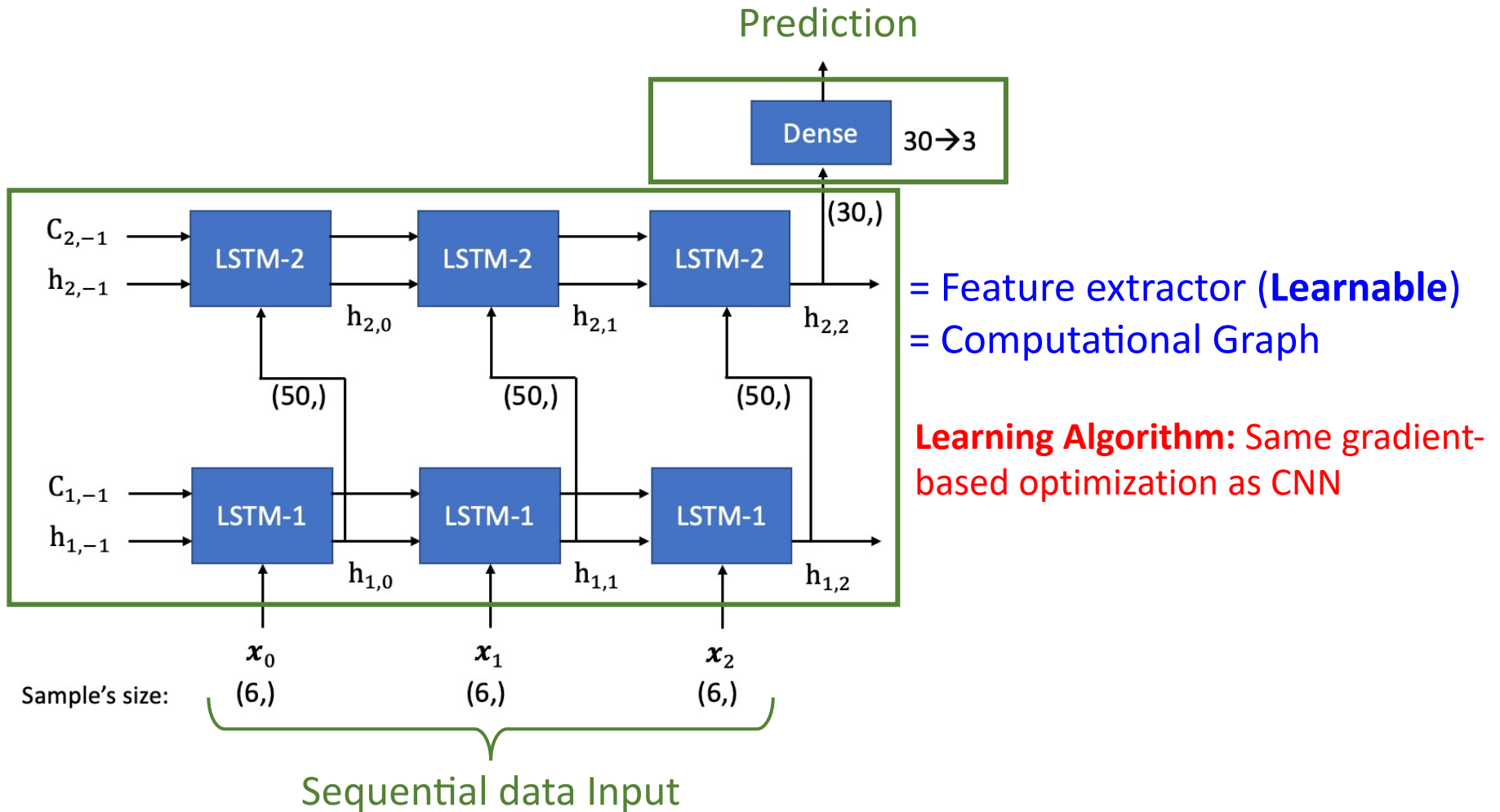
$\odot$ Elementwise operator

$\rightarrow$ Copy

$\rightarrow$ Concatenate



RNN $\rightarrow$ LSTM / GRU



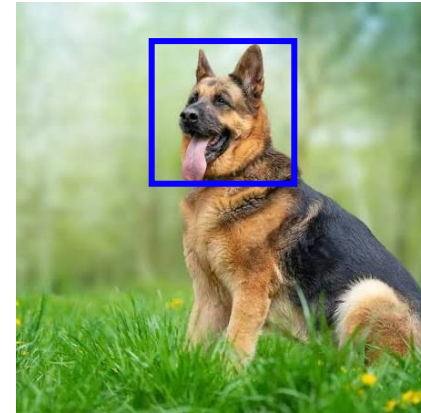
Learning Features: GNN

- From Grids & Sequences to Graphs
- Why Graph Neural Networks?
- GNN: Overall System Design



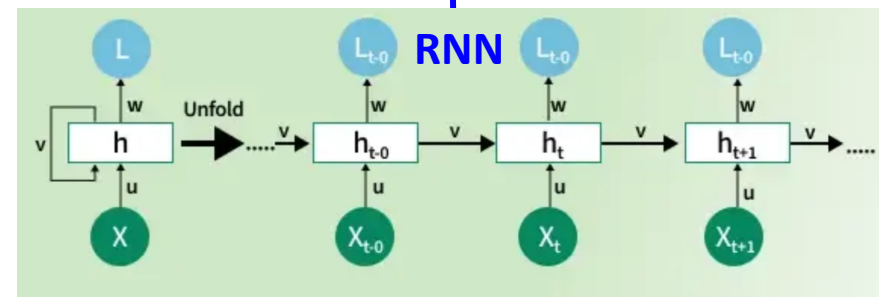
From Grids & Sequences to Graphs

- **CNNs**: Learn features from *grid-structured* data (images, audio spectrograms).
- **RNNs**: Learn features from *sequential* data (text, time series).
- Both assume **regular structures** (fixed neighbors, ordered data).



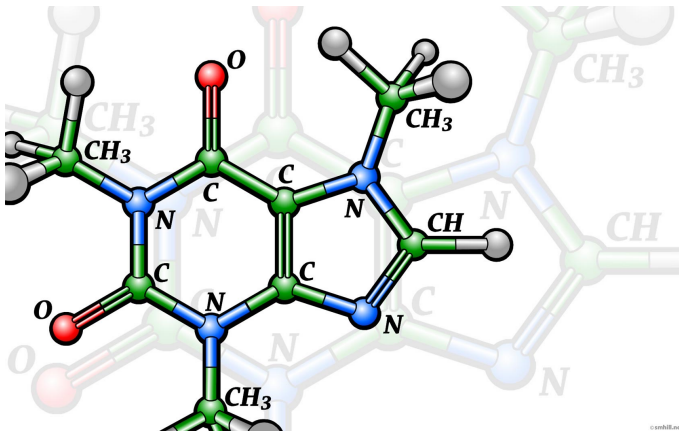
CNNs

“The picture contains a dog”



Why Graph Neural Networks?

- Need models that can:
 - Handle **arbitrary connections** (no fixed grid).
 - Preserve **permutation invariance** (node order irrelevant).
 - Propagate **information along edges**.
- Enter: **Graph Neural Networks (GNNs)**.

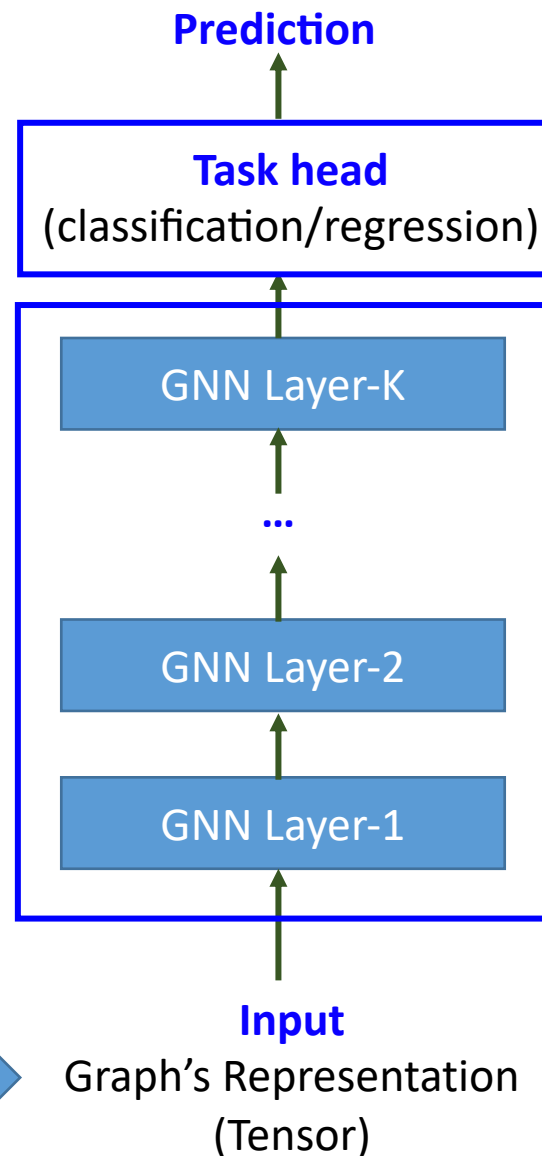
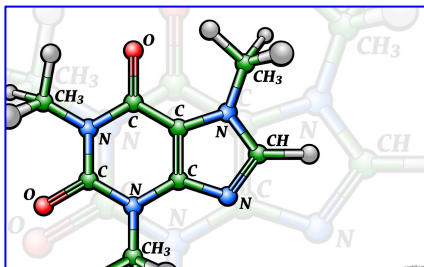




GNN: Overall System Design

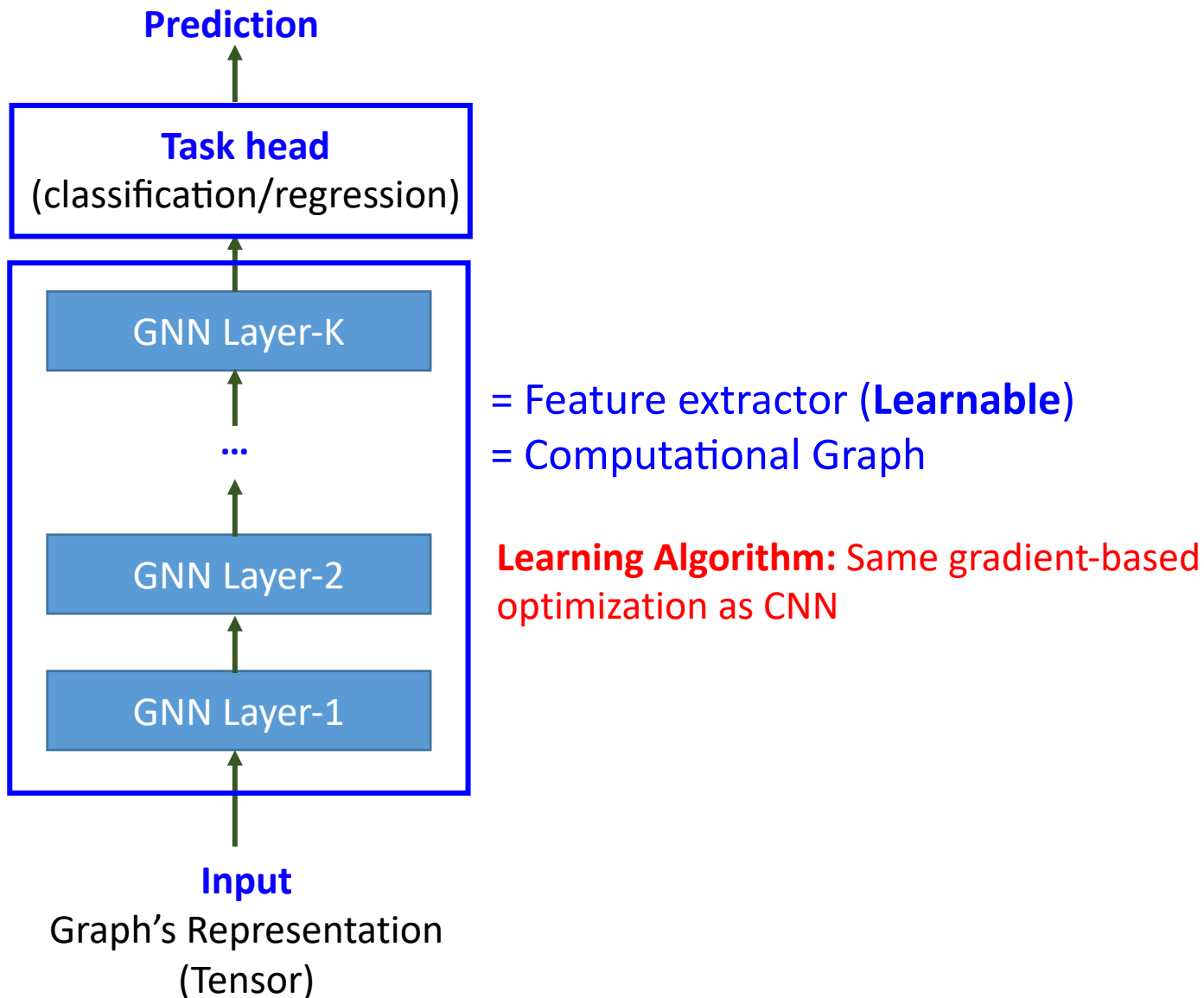
❖ Core Idea:

- Nodes = entities (e.g., atoms, users).
- Edges = relationships (e.g., chemical bonds, friendships).
- Message Passing:
 - ✓ Each node aggregates information from its neighbors.
 - ✓ Update rule:
 $\text{Node state} \leftarrow F(\text{self}, \text{neighbors})$.





GNN: Overall System Design

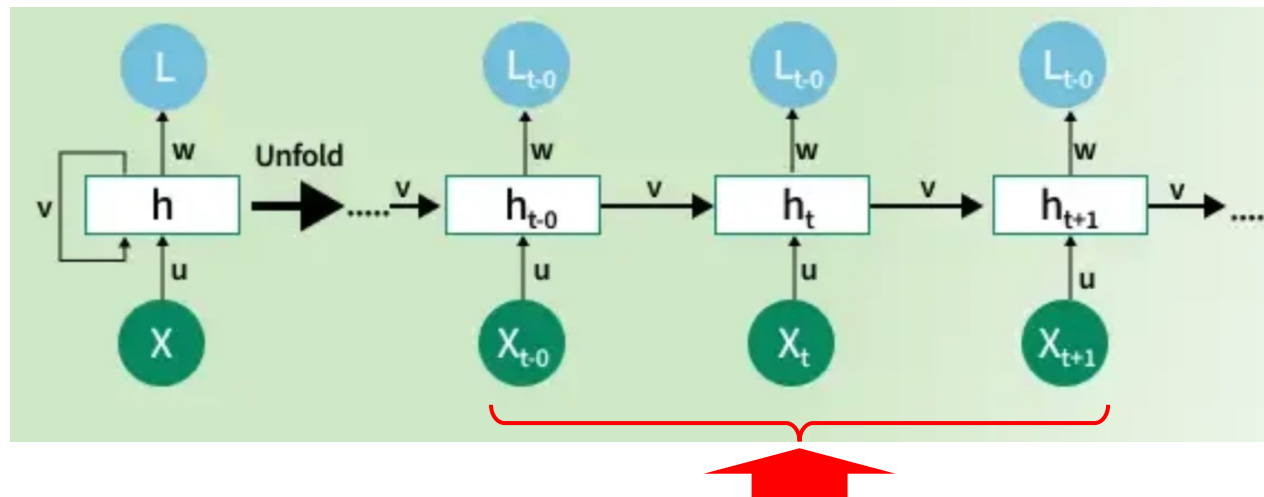


Learning Features: Transformer

- Why Transformers?
- Transformers: Encoder, Decoder, and Beyond
- Transformers: Encoder/Decoder
- From Text to Everything

Why Transformers?

- Limitations of RNNs: sequential bottleneck, vanishing gradients
- Need for parallelization and long-range dependency modeling

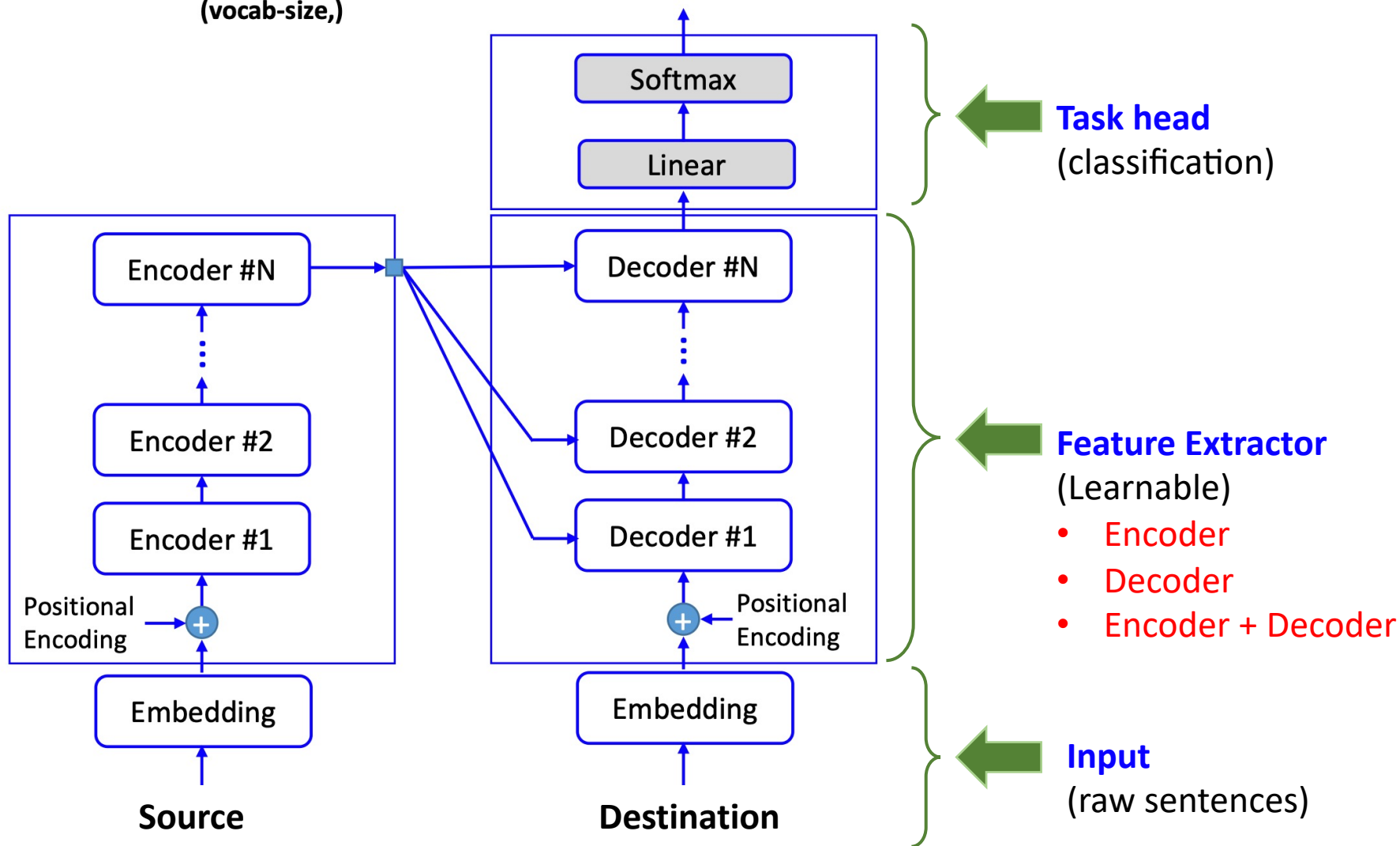


input samples:
supplied sequentially → really slow!



Transformers: Encoder, Decoder, and Beyond

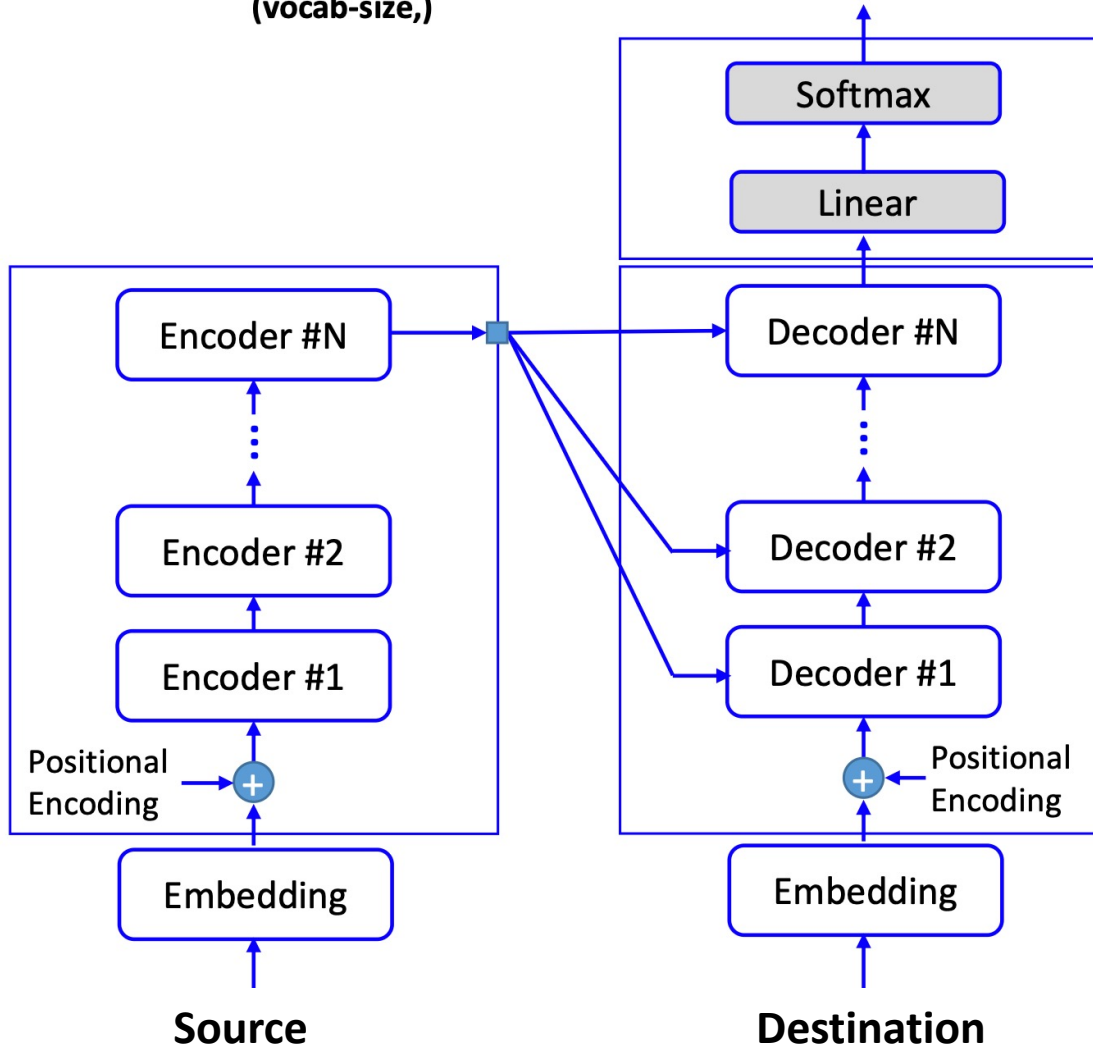
Conditional Distribution: $P(\text{next-word} \mid \text{Inputs, previous-predicted-words})$
(vocab-size,)





Transformers: Encoder, Decoder, and Beyond

Conditional Distribution: $P(\text{next-word} \mid \text{Inputs, previous-predicted-words})$
(vocab-size,)



A computational graph

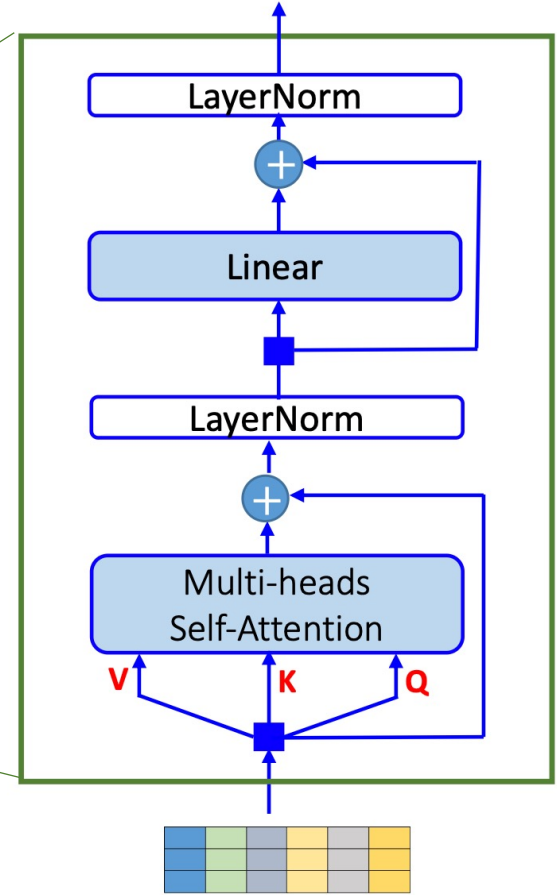
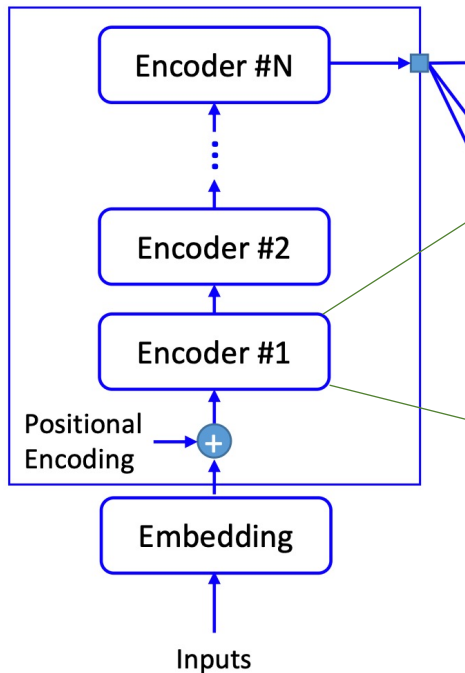
Learning Algorithm:
Same gradient-based
optimization as CNN



Transformers: Encoder/Decoder

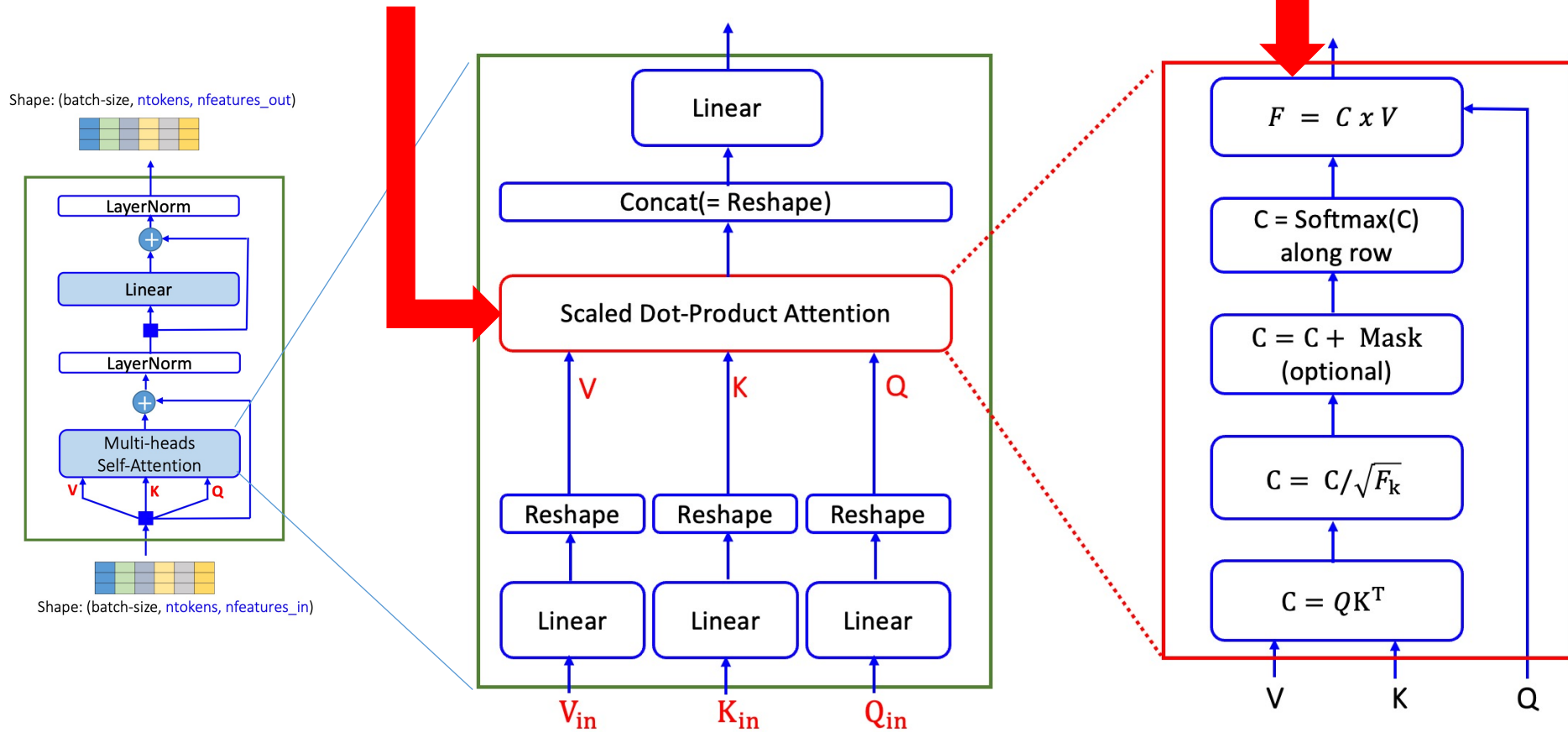
Shape: (batch-size, $ntokens$, $nfeatures_out$)

Encoder = sub-computational graph



Transformers: Encoder/Decoder

Attention mechanism





Transformers: Encoder/Decoder

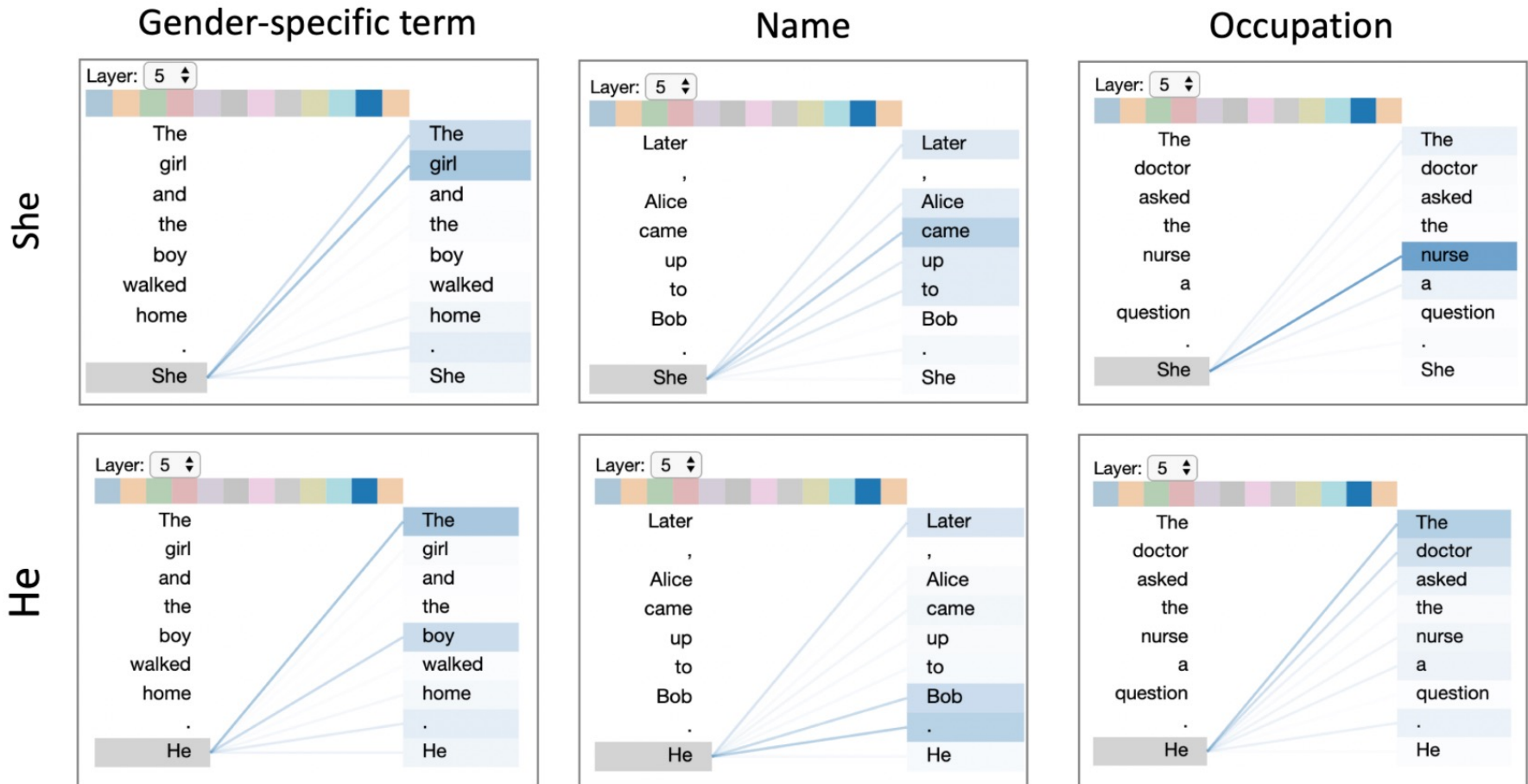


Figure 4: Attention pattern in GPT-2 related to coreference resolution suggests the model may encode gender bias.



From Text to Everything

Transformer (2017)

NLP:

- BERT
- GPT, ...

Vision:

- Backbone (Classification): ViT
- Object Detection: DETR
- Segmentation: SAM, ...

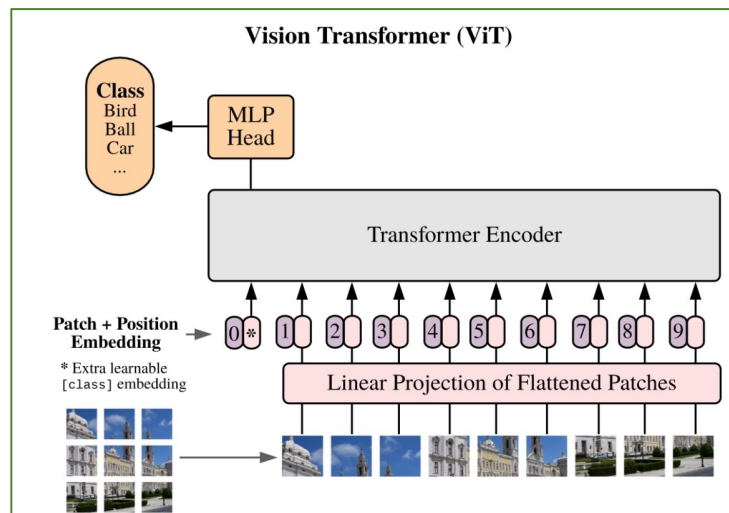
Audio:

- Whisper, ...

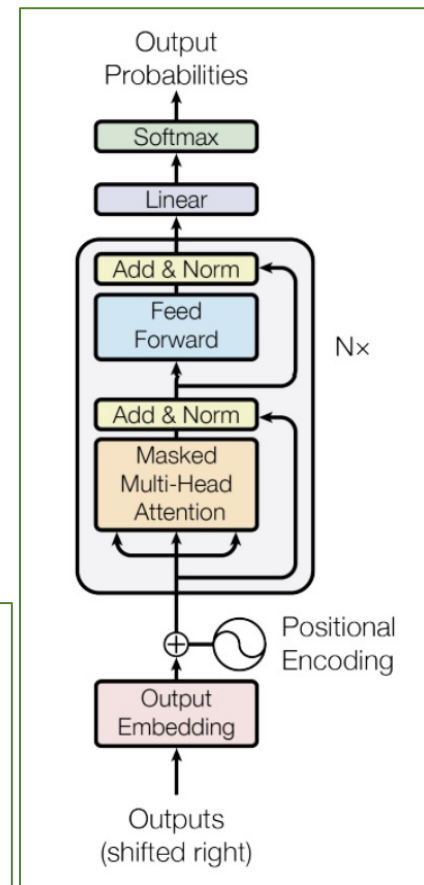
Multimodal:

- CLIP
- Flamingo
- GPT-4o

ViT



GPT



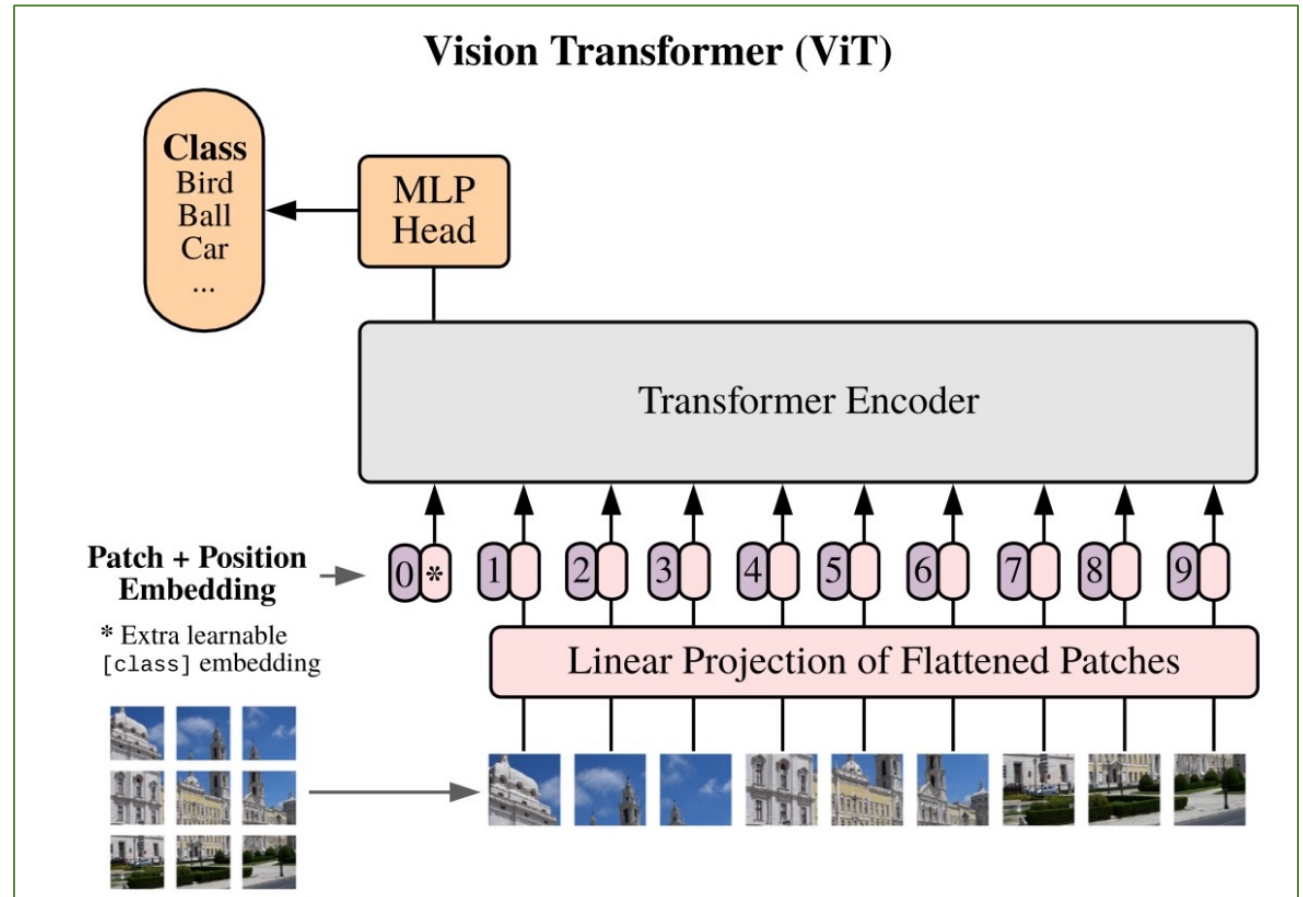
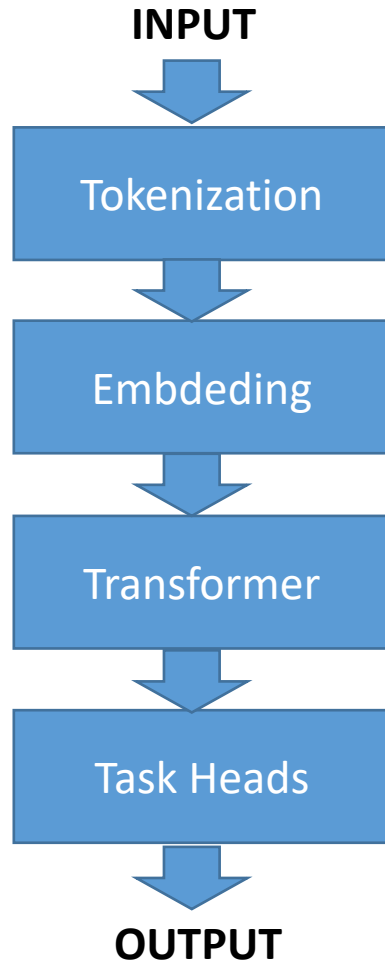


From Text to Everything: Principle





From Text to Everything: Principle



Learning Features: Mamba

- Limitations of Transformers
- State-Space Models: A New Paradigm

Limitations of Transformers

- ❖ Transformer excels at modeling complex interactions but suffers from:
 - **Quadratic time and memory complexity** (self-attention)
 - **Latency and inefficiency** in long-sequence tasks (e.g., audio, genomics, long chat history)
- ❖ Motivates the development of **linear-time sequence models**
- ❖ Recent shift: From **attention-based** to **state-space-based** architectures → **Mamba** is a breakthrough in this direction

	[CLS]	the	cat	chased	the	mouse	[SEP]	it	was	a	fast	chase	[SEP]	[PAD]
[CLS]	0.41	0	0.05	0	0.03	0.06	0.02	0.06	0.15	0.04	0.07	0.02	0.09	0
the	0	0	0	0.01	0.9	0	0.05	0.03	0	0	0	0	0	0
cat	0	0	0	1	0	0	0	0	0	0	0	0	0	0
chased	0	0	0	0	0.98	0	0.01	0.01	0	0	0	0	0	0
the	0.97	0	0	0	0	0	0	0	0.02	0	0	0	0	0
mouse	0	0.04	0	0.78	0.06	0	0.08	0.01	0	0	0	0.01	0	0
[SEP]	0.04	0.08	0.1	0.11	0.03	0.08	0.11	0.06	0.06	0.07	0.05	0.16	0.06	0
it	0.23	0	0.12	0	0	0.17	0.02	0.03	0.14	0.12	0.05	0.06	0.06	0
was	0.13	0.03	0.09	0.02	0.05	0.09	0.08	0.09	0.11	0.07	0.08	0.07	0.09	0
a	0	0.04	0	0.51	0.26	0	0.14	0.02	0	0	0.01	0.01	0.01	0
fast	0	0.09	0	0.67	0.19	0	0.02	0.01	0	0	0.01	0.01	0	0
chase	0.54	0	0	0	0.23	0	0	0.07	0.07	0	0.04	0	0.04	0
[SEP]	0.36	0	0.07	0	0.01	0.1	0.02	0.05	0.16	0.07	0.06	0.02	0.08	0
[PAD]	0.02	0.15	0.03	0.33	0.18	0.02	0.07	0.05	0.02	0.02	0.04	0.04	0.03	0



Quadratic time and memory complexity

$O(N^2)$

N: number of tokens

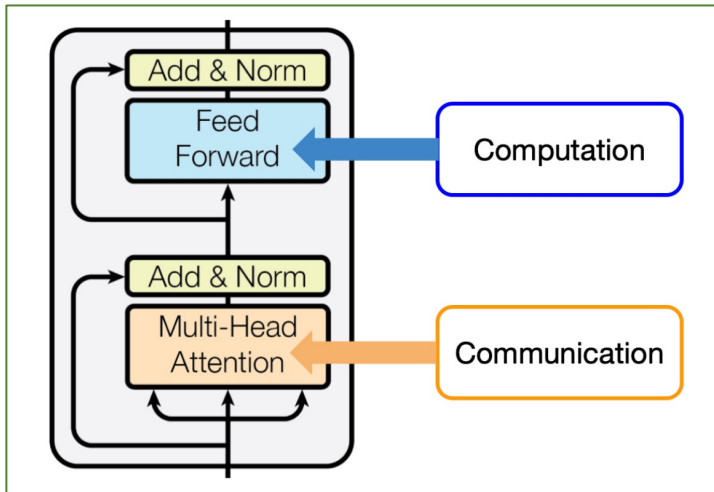


State-Space Models: A New Paradigm

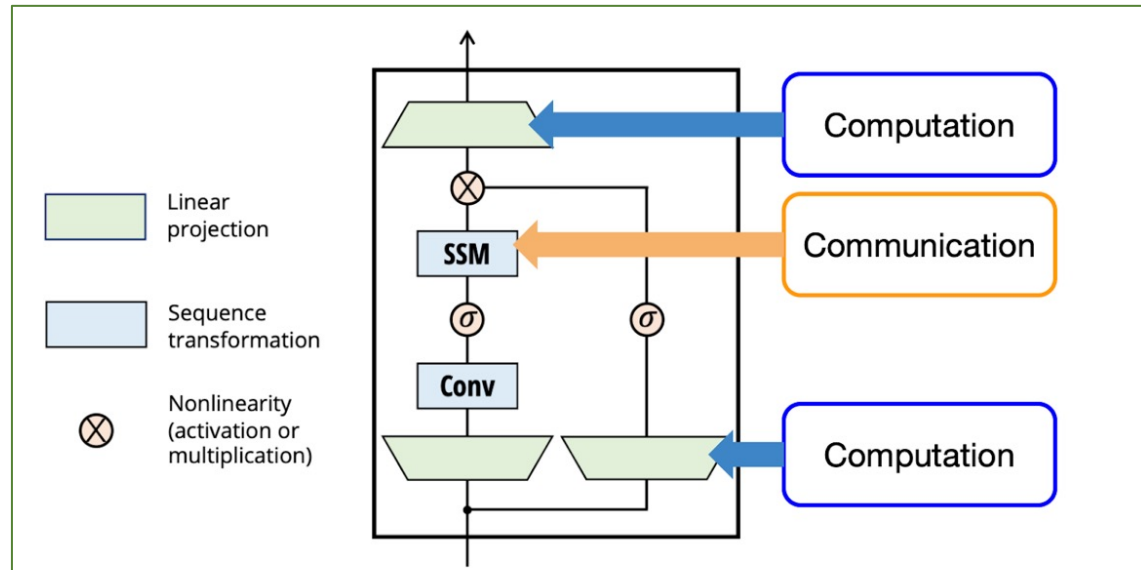
- ❖ SSMs describe sequence dynamics with **recurrence + convolution**
- ❖ Traditionally used in control systems and time-series
- ❖ Offer:
 - **Linear-time computation**
 - Long memory via implicit recurrence
 - Flexibility for different modalities (text, audio, genomics)



Transformer vs Mamba



The Transformer block



The Mamba block

- **Communication** *between* tokens
- **Computation** *within* a token

The Transformer block or The Mamba block
= A sub-computation graph

Learning Algorithm: Same gradient-based optimization as CNN



Transformer vs Mamba

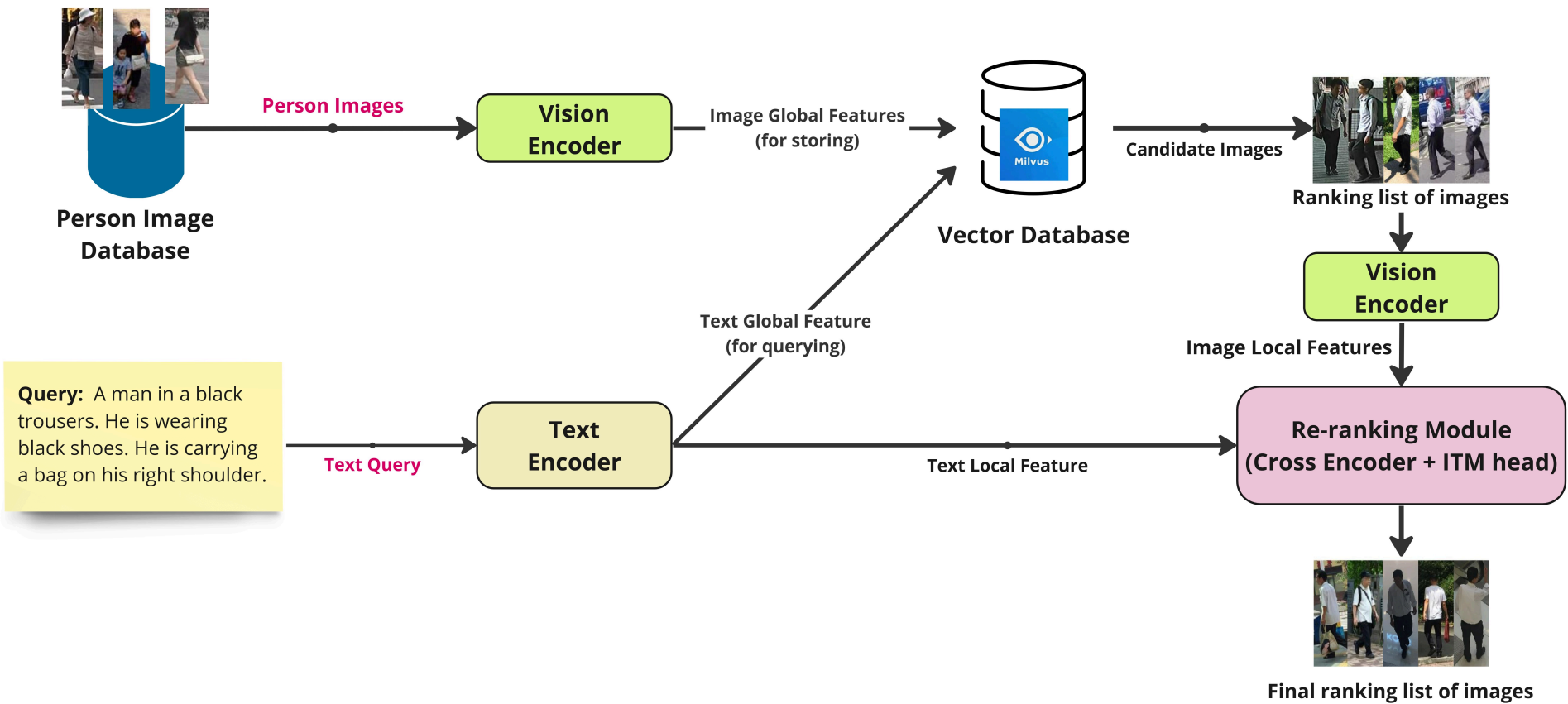
Feature	Transformer	Mamba
Core Mechanism	Self-attention	State-space recurrence
Complexity	$O(N^2)$	$O(N)$
Memory Span	Soft-attention limit	Long-range memory
Inference Efficiency	Slow (step-by-step)	Fast (streaming-friendly)
Parallel Training	Yes	Yes
Modality Support	NLP, vision, audio	NLP, audio, genomics

Applications of Feature Learning in Multimodal AI

- Text-Based Human Search
- Spatial Intelligence Systems



Text-Based Human Search





Text-Based Human Search

Query: “nam đội bảo hiểm, mặc áo xanh lá”

The screenshot shows a video search interface with a dark blue background. At the top, there is a search bar with the query “nam đội bảo hiểm, mặc áo xanh lá” highlighted by a red oval. To the left of the search bar is a “Tất cả camera” button, and to the right is a “Độ chính xác” slider. Below the search bar is a timeline for the date 25/03, showing a sequence of frames with numbers in circles: 00 (110), 01 (106), 02 (110), 03 (110), 04 (97), 05 (3), 06 (12), 07 (3), 08 (53), 09 (188), 10 (230), 11 (232), 12 (97), 13 (7), 14, 15, 16, 17, 18, 19, 20, 21, 22, 23. Below the timeline are three buttons: “Hình ảnh”, “Playback”, and “Camera”. Under the “Hình ảnh” button, it says “Camera mới” and “8891 kết quả”. Below this is a grid of 20 small video frames showing a person in a green shirt riding a motorcycle. At the bottom left, there is a link “Xem thêm”.

Trở về

Tất cả camera

25/03 - 25/03 | 00:00 - 23:59 | nam đội bảo hiểm, mặc áo xanh lá

Độ chính xác

25/03

00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23

110 106 110 110 97 3 12 3 53 188 230 232 97 7

Hình ảnh Playback Camera

Camera mới

8891 kết quả

Results: image + camera (location) + time

Xem thêm



Text-Based Human Search



Caption 1: The person in the image is viewed from behind, wearing a black hooded sweatshirt with a white shirt underneath, light-washed denim shorts, black leggings, and white sneakers. The person appears to be carrying a large black backpack with multiple pockets and compartments. The person's head is covered by the hood, and their face is not visible. The person's arms are bent at the elbows, with their hands tucked into their pockets. The person is standing on a sidewalk or street, with a wall in the background.

Caption 2: The person in the image appears to be wearing a white long-sleeved shirt with a black design on the front. They have on black leggings and white sneakers. The individual is also carrying a black backpack with white accents. The hairstyle is not clearly visible, but it seems to be a short cut. The gender of the person is not definitively identifiable from the image.



Caption 1: The person in the image is wearing a plaid shirt with a mix of blue and white colors. They have on dark jeans and appear to be carrying a brown leather bag over their shoulder. The individual is wearing a face mask, which covers the lower part of their face. Their hairstyle is short and appears to be neatly trimmed. The gender of the person is not clearly identifiable from the image.

Caption 2: The man has short, light brown hair and is wearing a blue and white plaid shirt, dark blue jeans, and a brown leather bag slung over his shoulder. He is also wearing a face mask.

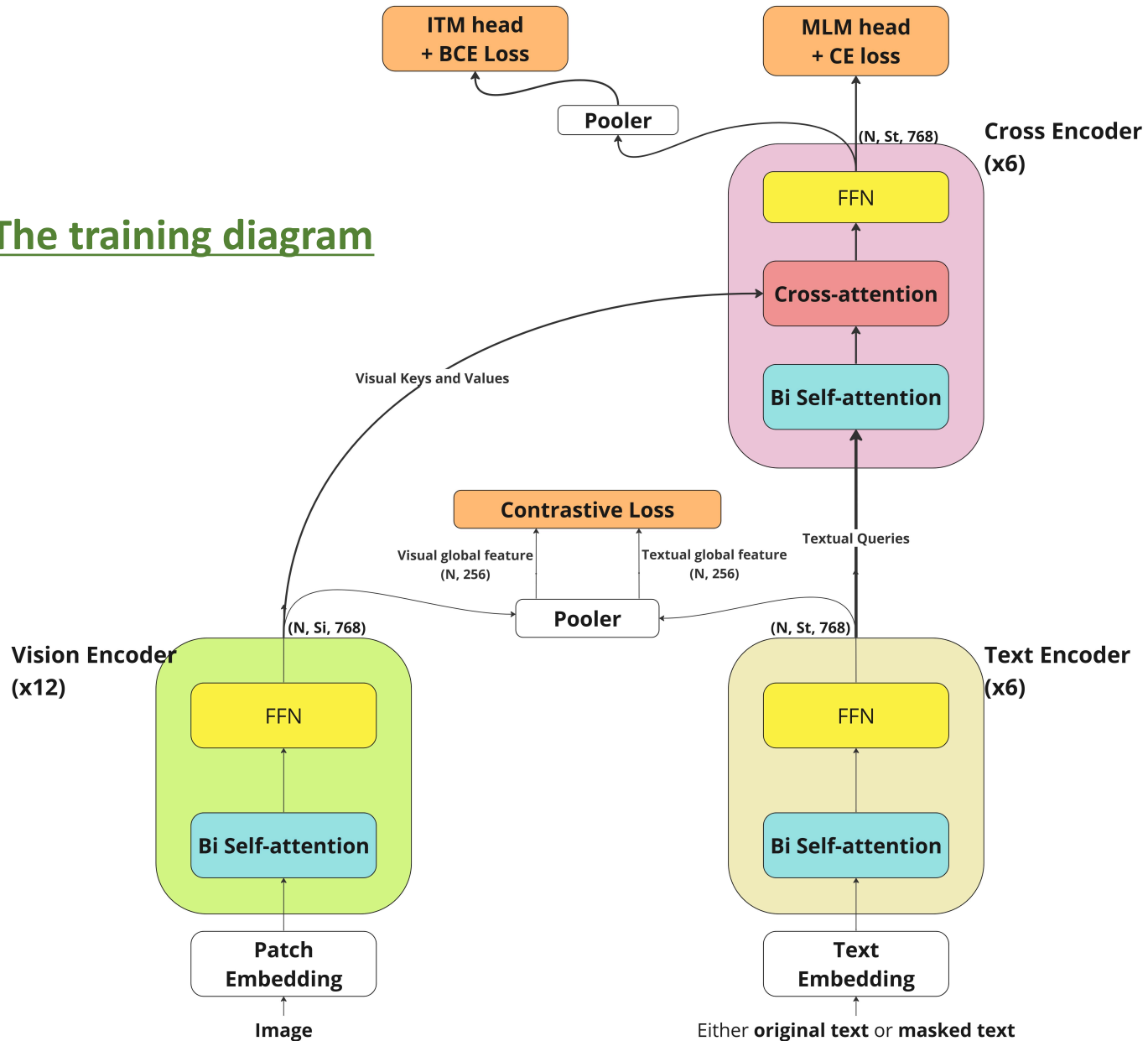
Datasets: organize in triples (person-id, image, caption)

- **For each person:**
 - ✓ Assign a unique identity (an integer)
 - ✓ Capture some images (as shown above)
- **For each image:**
 - ✓ Provide several captions describing the person in the image



Text-Based Human Search

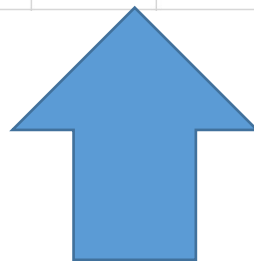
The training diagram





Text-Based Human Search

Methods	Conference/Journal (2023/2024)	Datasets/Metrics											
		CUHK				RSTP				ICFG			
		R1	R5	R10	mAP	R1	R5	R10	mAP	R1	R5	R10	mAP
IRRA	CVPR 23	73,38	89,93	93,71	66,13	60,2	81,3	88,2	47,17	63,46	80,25	85,82	38,06
TBPS-CLIP	AAAI 24	73,54	88,19	92,35	65,38	61,95	83,55	88,75	48,26	65,05	80,34	85,47	39,83
APTМ	ACM MM 23	76,53	90,04	94,15	66,91	67,5	85,7	91,45	52,56	68,51	82,99	87,56	41,22
RaSa	ICJAI 23	76,51	90,29	94,25	69,38	66,9	86,5	91,35	52,31	65,28	80,4	85,12	41,29
UniPT	ICCV 23	68,5	84,67	90,38	-	51,85	74,85	82,85	-	60,09	76,19	82,46	-
RDE	CVPR 24	75,94	90,63	94,04	67,92	65	84,75	90,6	51,57	67,6	82,47	87,17	40,34
LCR2S	Mm 23	67,36	84,19	89,62	59,24	54,95	76,65	84,7	40,92	57,93	76,08	82,4	38,21
CFAM	CVPR 24	76,71	91,83	95,96	68,47	63,54	84,75	92,32	50,48	66,58	82,47	87,37	40,46
BiLMa	ICCV 23	74,03	89,59	93,62	66,57	61,2	81,5	88,8	48,51	63,83	80,15	85,74	38,26
Ours		77,92	91,85	95,53	70,12	69,55	84,85	91,35	54,56	71,2	84,87	89,34	44,26
Notes:	Red: 1st, Blue: 2nd, Pink: 3rd												

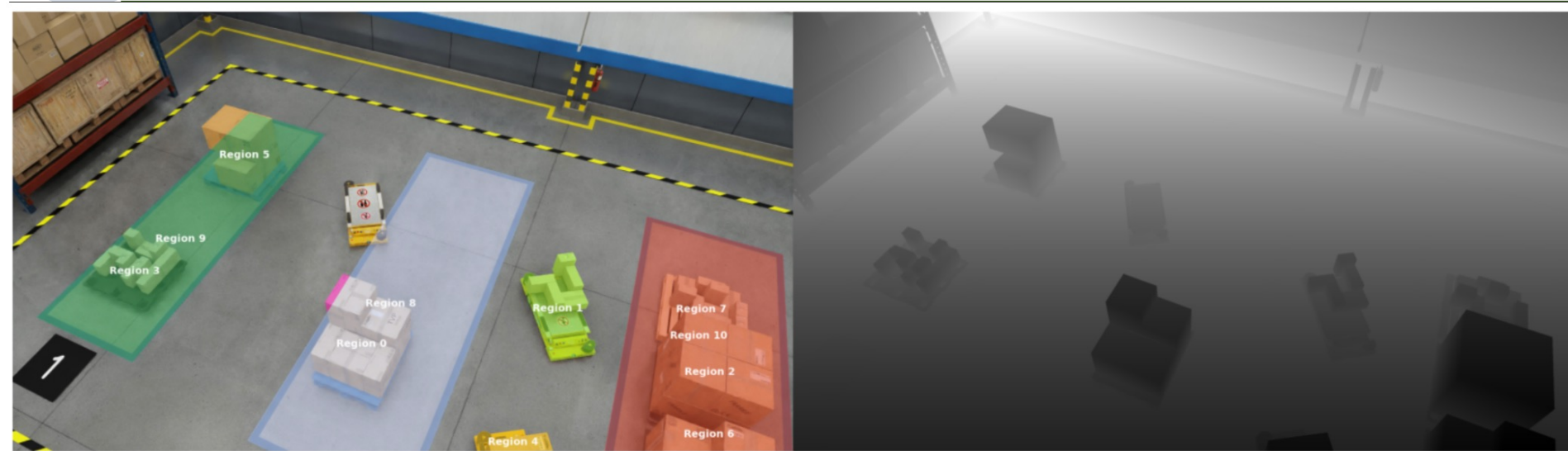


We achieved mostly top-1 to top-3 results across various benchmarks.



Spatial Intelligence Systems

The problem



Q: Considering the pallets <mask> <mask> <mask> <mask> <mask> <mask> <mask> <mask>, what is the total number of pallets in the rightmost buffer region among <mask> <mask> <mask>?

GT: Among the buffer regions, the buffer region [Region 10] is the rightmost buffer region. The buffer region [Region 10] contains pallets [Region 2] [Region 6] [Region 7]. Hence, in buffer area [Region 10], there are exactly three pallets.

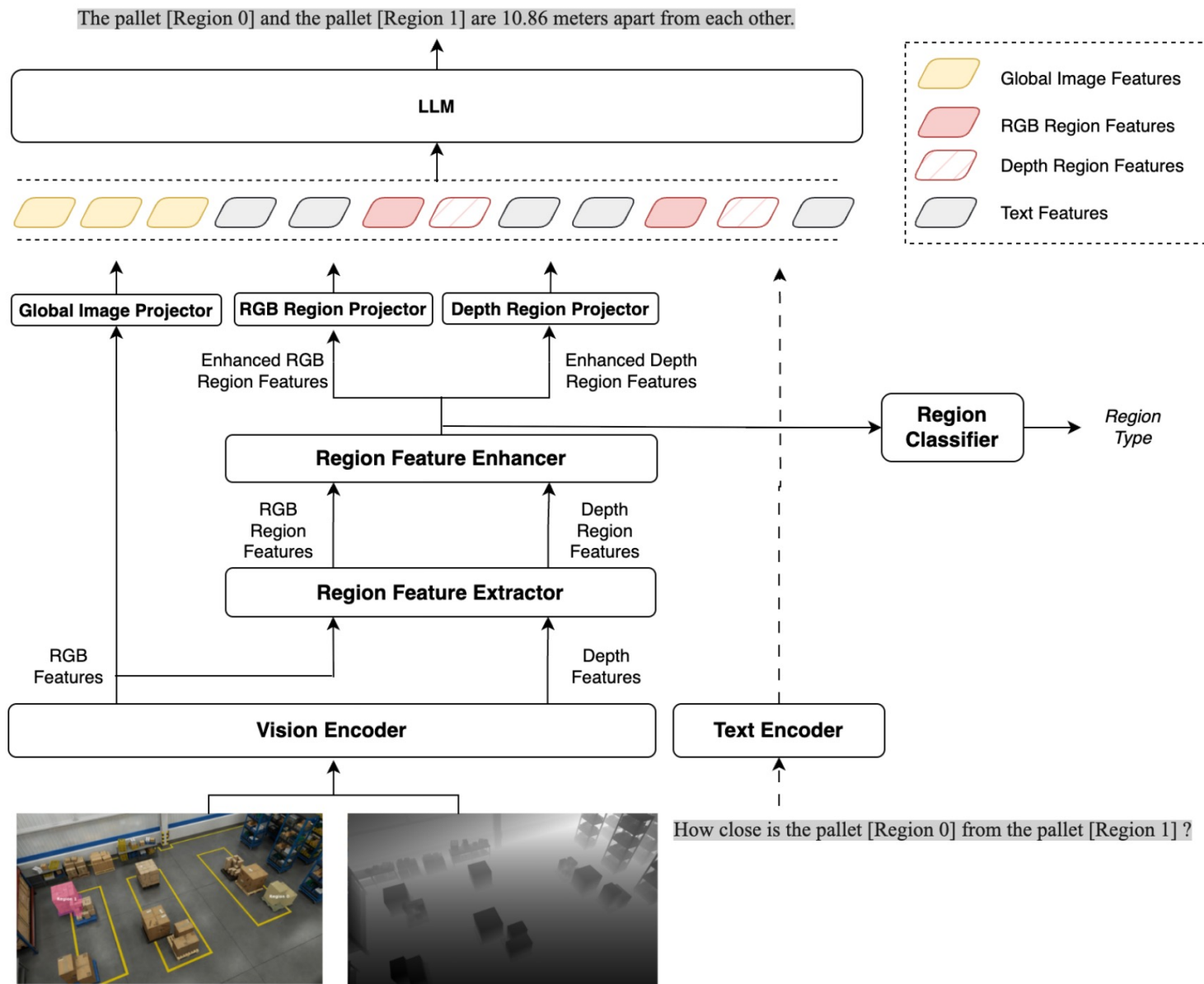
Spatial Intelligence Systems:

- Inputs:
 - An Image (RGB)
 - A Depth Image (Grayscale)
 - Q: A Question (text), as shown above
- Output:
 - GT: Ground-Truth (text), the expected answer, as shown above



Spatial Intelligence Systems

The overall architecture





Q: Considering the pallets <mask> <mask> <mask> <mask> <mask> <mask> <mask> <mask>, what is the total number of pallets in the rightmost buffer region among <mask> <mask> <mask>?

GT: Among the buffer regions, the buffer region [Region 10] is the rightmost buffer region. The buffer region [Region 10] contains pallets [Region 2] [Region 6] [Region 7]. Hence, in buffer area [Region 10], there are exactly three pallets.

Baseline: The buffer region [Region 10] is the rightmost buffer region among all the buffer regions. The buffer region [Region 10] has pallets [Region 2] [Region 7]. Therefore, the buffer area [Region 10] has two pallets.

Phase 4: The buffer region [Region 10] is the rightmost buffer region among all the buffer regions. The buffer region [Region 10] has pallets [Region 2] [Region 6] [Region 7]. Therefore, the buffer region [Region 10] holds a total of three pallets.



Spatial Intelligence Systems

Some results



Q: Measure the distance from the leftmost pallet to the shelf <mask> among the pallets <mask> <mask> <mask> <mask> <mask>.

GT: [Region 5] is the leftmost pallet from this viewpoint. The distance from the pallet [Region 5] to the shelf [Region 0] is approximately 13.47 meters.

Baseline: [Region 5] is the leftmost pallet among all the pallets. The distance between the pallet [Region 5] and the shelf [Region 0] is 9.8 meters.

Phase 4: [Region 5] is the leftmost pallet among all the pallets. The distance between the pallet [Region 5] and the shelf [Region 0] is 13.42 meters.



Spatial Intelligence Systems

Some results

Table 2. Validation results across training stages.

Stage	Count (Acc@10)	Dist. (Acc@10)	L/R (Acc)	MCQ (Acc)	Overall (%)
Baseline	88.60	80.86	99.80	94.52	90.95
Phase 1	93.80	84.57	99.80	93.86	93.01
Phase 2	94.60	85.39	99.80	96.05	93.98
Phase 4	95.40	85.39	99.80	96.71	94.34



Spatial Intelligence Systems

Some results

Leaderboard

Refresh Frequency 300 seconds

Track 1

Track 2

Track 3

Track 4

Warehouse Spatial Intelligence

General



Rank	Team ID	Team Name	Score
1	16	UWIPL_ETRI	96.0789
2	57	HCMUT.VNU	91.9735
3	140	Embia	90.6772
4	49	MIZSU	73.0606
5	99	HCMUS_HTH	66.8861
6	36	MealsRetrieval	56.6352
7	183	BKU22	50.3662
8	67	Smart Lab	31.9245
9	219	AICV	28.2993

We ranked 2nd in the AI-CITY CHALLENGE 2025.



The Future of Feature Learning



From Handcrafted to Learned

- Early AI: SIFT, HOG (manual features)
- Deep learning: CNNs, RNNs, GNNs, Transformers



Multimodal & Generalizable

- Transformers enable cross-domain features
- CLIP, Flamingo: joint vision-language-audio
- Embeddings now reusable across tasks



Next: Efficiency & Long Context

- Transformer = slow on long sequences
- New models (Mamba, RWKV, Hyena)
- Shift: **Deep** → **Efficient & Contextual**



Conclusion

✓ What We've Achieved

- Unified **representation learning** across text, image, audio, and graphs.
- Transformer became the **foundation model** for general-purpose AI.

⚠ What's Holding Us Back

- Latency & inefficiency for **long sequences**.
- Trade-off between **depth, context, and compute**.

🚀 Where We're Going

- New models (Mamba, RWKV, Hyena) shift from “attention” to **continuous state space**.
- Future = **Efficient, Contextual, and Multimodal Learning**
→ For AI to understand **long conversations, genomic sequences, videos, and lifelong memory**.



Thank you for your attention!

Looking forward to your thoughts 

Thanh-Sach LE
LTSACH@hcmut.edu.vn